

DIRECT CONTROL SERVICES

ANNUAL PRICING PROPOSAL 2012/13

.....
30 APRIL 2012
.....

DELIVERING EFFICIENT AND
SUSTAINABLE NETWORK SERVICES



**Endeavour
Energy**

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Appendices:

Item No.	Description
A	CONFIDENTIAL – Distribution Use of System Tariffs
B	CONFIDENTIAL – Consideration of LRMC
C	CONFIDENTIAL – Climate Change Fund Recovery Tariffs
D	CONFIDENTIAL – Transmission Use of System Tariffs
E	CONFIDENTIAL – Network Use of System Tariffs
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H	Public Lighting (Rates 3 and 4)
I	Time of Day and Season Definitions
J	CEG Review – Regulatory Cost of Supply Methodology
K	CEG Review – Long Run Marginal Cost Methodology

Attachments:

Item No.	Description
1	CONFIDENTIAL – AER Weighted Average Price Cap Compliance Model
2	CONFIDENTIAL – Long Run Marginal Cost Model
3	CONFIDENTIAL – Regulatory Cost of Supply Model
4	CONFIDENTIAL – Public Lighting Model (Tariff Classes 1 and 2)
5	CONFIDENTIAL – Public Lighting Model (Tariff Classes 3 and 4)

INTRODUCTION

On 1 March 2011, Endeavour Energy (formerly known as Integral Energy Australia) completed the sale of its retail business (including the name Integral Energy Australia) to Origin Energy as part of the NSW Government's energy reform strategy. All references to Integral Energy in this *2012/13 Annual Pricing Proposal* are references to Endeavour Energy.

Continuing real increases in electricity prices are placing a considerable strain on family budgets. Endeavour Energy is continuing to pursue a number of initiatives to reduce costs and improve productivity to moderate future network tariff increases.

Endeavour Energy is submitting this *2012/13 Annual Pricing Proposal* to the Australian Energy Regulator (AER) in accordance with the requirements of the National Electricity Rules (the *Rules*), particularly clause 11.15.2 of the Rules and Appendix 1 of Chapter 11, an amended form of Chapter 6 of the Rules applicable to the NSW and ACT distribution businesses, for the purposes of the 2009-2014 regulatory control period (the *Transitional Rules*).

1.1 National Electricity Rules

In accordance with clause 6.1.1 of the *Transitional Rules*, the AER is responsible for the regulation of distribution services provided by means of, or in connection with, distribution systems that form part of the national grid.

1.1.1 Classification of distribution services

In accordance with clause 6.2.3A of the *Transitional Rules*, Distribution services to be provided by the Distribution Network Service Provider (DNSP) are divided into the following three classes:

1. Direct control services;
2. Negotiated distribution services; and
3. Unregulated distribution services.

Direct control services are further divided into the following two subclasses:

1. Standard control services; and
2. Alternative control services.

This pricing proposal is relevant to those services provided by Endeavour Energy that are classified as direct control services.

1.2 Scope of Endeavour Energy's pricing proposal

In meeting the requirements of Part I of the *Transitional Rules*, and in particular clause 6.18.2(a)(2), Endeavour Energy's *2012/13 Annual Pricing Proposal* sets out the proposed prices required to fund the capital and operating investment programs approved by the Australian Energy Regulator (AER) in the *2009 Distribution Determination* to ensure a safe and reliable network over the *2009 regulatory control period*.

This *2012/13 Annual Pricing Proposal* is submitted in accordance with, and complies with, the requirements of:

1. The National Electricity Law;
2. The National Electricity Rules including the *Transitional Rules*;
3. The AER's 2009-10 to 2013-14 New South Wales distribution determination (*2009 Distribution Determination*), as adjusted for the outcomes of the Australian Competition Tribunal's review;
4. The Australian Competition Tribunal's Merits Review decision; and
5. The AER's Final Decision on control mechanisms for direct control services for the ACT and NSW 2009 distribution determination.

1.3 Structure of Endeavour Energy's pricing proposal

Endeavour Energy's 2012/13 Annual Pricing Proposal is structured as follows:

Table 1.1 – Structure of Endeavour Energy's pricing proposal

Chapter	Title	Purpose
2	Regulatory Requirements	Summarises the regulatory requirements as they relate to Endeavour Energy's <i>pricing proposal</i> , and includes the relevant requirements of the <i>Rules</i> and the AER's 2009 <i>Distribution Determination</i> .
3	Business Characteristics	Summarises the characteristics of Endeavour Energy's network that provide the context for Endeavour Energy's network tariff strategy.
4	Tariff Classes	Defines the tariff classes into which Endeavour Energy's customers for direct control services are divided.
5	Network Tariff Strategy	Outlines Endeavour Energy's Network Tariff Strategy.
6	Network Use of System Tariffs	Describes the nature and extent of the change in Endeavour Energy's NUOS tariffs between 2011/12 and 2012/13.
7	Customer Impacts	Outlines the expected customer impacts of Endeavour Energy's pricing proposal.
8	Compliance with Regulatory Requirements	Demonstrates that Endeavour Energy's pricing proposal complies the regulatory requirements as they relate to Endeavour Energy's annual pricing proposal.
9	Climate Change Fund	Sets out the amount that Endeavour Energy must recover from its Climate Change Fund Recovery tariffs.
10	Transmission Cost Recovery Tariffs	Sets out Endeavour Energy's Transmission cost recovery tariff setting methodology and demonstrates adjustments made to the tariffs resulting from the operation of the transmission use of system overs and unders account.
11	Customer Reassignment	Sets out the number of customers impacted by and reasons for Endeavour Energy's 2012/13 tariff reassignment strategy
12	Alternate Control Services – Public Lighting Tariffs	Sets out Endeavour Energy's Alternate Control Service – Public Lighting Tariffs in accordance with the 2009 <i>Distribution Determination</i> .
Appendices		Separately provided

1.4 Confidentiality

Clause 6.19.2(a) of the *Transitional Rules* provides that “all information about a Service Applicant or Distribution Network User used by Distribution Network Service Providers for the purposes of distribution service pricing is confidential information.”

The following appendices and attachments to the *2012/13 Annual Pricing Proposal* contain sensitive confidential information specific to the individual *Distribution Network Users*. As such, Endeavour Energy requests that the AER does not disclose the information contained in these appendices and attachments to any person except as permitted by the Law and Rules.

Appendices

1. Confidential Appendix A – Distribution Use of System Tariffs;
2. Confidential Appendix B – Consideration of LRMC;
3. Confidential Appendix C – Climate Change Fund Recovery Tariffs;
4. Confidential Appendix D – Transmission Cost Recovery Tariffs;
5. Confidential Appendix E – Network Use of System Tariffs; and
6. Confidential Appendix F – Public Lighting (Rates 1 and 2).

Attachments

1. Confidential Attachment 1 – AER Weighted Average Price Cap (WAPC) Compliance Model;
2. Confidential Attachment 2 – Long Run Marginal Cost Model;
3. Confidential Attachment 3 – Regulatory Cost of Supply Model;
4. Confidential Attachment 4 – Public Lighting Model (Tariff Classes 1 and 2); and
5. Confidential Attachment 5 – Public Lighting Model (Tariff Classes 3 and 4).

2

REGULATORY REQUIREMENTS

This Chapter summarises the regulatory requirements as they relate to Endeavour Energy's 2012/13 *Annual Pricing Proposal*, and includes the relevant requirements of the *Rules* and the AER's 2009 *Distribution Determination*.

2.1 Requirements of the Rules

2.1.1 Required elements of a pricing proposal

In accordance with clause 6.18.2 of the *Transitional Rules*, Endeavour Energy's 2012/13 *Annual Pricing Proposal* must include the elements summarised below:

6.18.2 Pricing proposals

(b) A pricing proposal must:

- (1) set out the tariff classes that are to apply for the relevant regulatory year; and
- (2) set out the proposed tariffs for each tariff class; and
- (3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates; and
- (4) set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year; and
- (5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur; and
- (5A) in the case of a NSW Distribution Network Service Provider – set out how Climate Change Fund payments are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those amounts in any previous regulatory year; and

(6) set out how designated pricing proposal charges are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year; and

(7) demonstrate compliance with the Rules and any applicable distribution determination; and

(8) describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination.

(b1) In its first year pricing proposal after 1 July 2010 the NSW Distribution Service Provider must detail how the provider proposes to carry out adjustments to Climate Change Fund payments for the purposes of clause 6.18.2(b)(5A) and report to the AER on the recovery process under clause 6.18.7B(a) to (c).

(c) The AER must on receipt of a pricing proposal from a Distribution Network Service Provider publish the proposal.

Endeavour Energy has prepared its *2012/13 Annual Pricing Proposal* in accordance with the requirements of the *Transitional Rules*, as summarised above.

In accordance with clause 6.18.2 of the *Transitional Rules*, Endeavour Energy must submit to the AER at least two months before the start of the second and each subsequent year of the *2009 regulatory control period*, a pricing proposal (annual pricing proposal) for the relevant year. Submission of the *2012/13 Annual Pricing Proposal* on 30 April 2012 satisfies this obligation.

2.1.2 Tariff Classes

In accordance with clause 6.18.3 of the *Transitional Rules*, Endeavour Energy must satisfy the following criteria with respect to tariff classes:

6.18.3 Tariff classes

(a) A pricing proposal must define the tariff classes into which customers for direct control services are divided.

(b) Each customer for direct control services must be a member of 1 or more tariff classes.

(c) Separate tariff classes must be constituted for customers to whom standard control services are supplied and customers to whom alternative control services are supplied (but a customer for both standard control services and alternative control services may be a member of 2 or more tariff classes).

(d) A tariff class must be constituted with regard to:

(1) the need to group customers together on an economically efficient basis; and

(2) the need to avoid unnecessary transaction costs.

2.1.3 Assignment or re-assignment of customers to tariff classes

In accordance with clause 6.18.4 of the *Transitional Rules*, Endeavour Energy is governed by the following customer reassignment principles:

6.18.4 Principles governing assignment or re-assignment of customers to tariff classes and assessment and review of basis of charging

(a) In formulating provisions of a distribution determination governing the assignment of customers to tariff classes or the re-assignment of customers from one tariff class to another, the AER must have regard to the following principles:

(1) customers should be assigned to tariff classes on the basis of one or more of the following factors:

(i) the nature and extent of their usage;

(ii) the nature of their connection to the network;

(iii) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement;

(2) customers with a similar connection and usage profile should be treated on an equal basis;

(3) however, customers with micro-generation facilities should be treated no less favourably than customers without such facilities but with a similar load profile;

(4) a Distribution Network Service Provider's decision to assign a customer to a particular tariff class, or to re-assign a customer from one tariff class to another should be subject to an effective system of assessment and review.

Note:

If (for example) a customer is assigned (or reassigned) to a tariff class on the basis of the customer's actual or assumed maximum demand, the system of assessment and review should allow for the reassignment of a customer who demonstrates a reduction or increase in maximum demand to a tariff class that is more appropriate to the customer's load profile.

(b) If the charging parameters for a particular tariff result in a basis of charge that varies according to the usage or load profile of the customer, a distribution determination must contain provisions for an effective system of assessment and review of the basis on which a customer is charged.

2.1.4 Pricing Principles

Clause 6.18.5 of the *Transitional Rules* state that Endeavour Energy must comply with the following pricing principles:

6.18.5 Pricing principles

(a) For each tariff class, the revenue expected to be recovered should lie on or between:

(1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and

(2) a lower bound representing the avoidable cost of not serving those customers.

(b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:

(1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and

(2) must be determined having regard to:

(i) transaction costs associated with the tariff or each charging parameter; and

(ii) whether customers of the relevant tariff class are able or likely to respond to price signals.

(c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

2.1.5 Recovery of charges for transmission use of system services

In accordance with clause 6.18.7 of the *Transitional Rules*, Endeavour Energy's 2012/13 *Annual Pricing Proposal* must satisfy the following criteria:

6.18.7 Recovery of designated pricing proposal charges

(a) A pricing proposal must provide for tariffs designed to pass on to customers the designated pricing proposal charges to be incurred by the Distribution Network Service Provider.

(b) The amount to be passed on to customers for a particular regulatory year must not exceed the estimated amount of the designated pricing proposal charges adjusted for over or under recovery in accordance with paragraph (c).

(c) The over and under recovery amount must be calculated in a way that:

(1) subject to subparagraphs (2) and (3) below, is consistent with the method determined by the AER in the relevant distribution determination for the Distribution Network Service Provider;

(2) ensures a Distribution Network Service Provider is able to recover from customers no more and no less than the designated pricing proposal charges it incurs; and

(3) adjusts for an appropriate cost of capital that is consistent with the rate of return used in the relevant distribution determination for the relevant regulatory year.

(d) Notwithstanding anything else in this clause 6.18.7, a Distribution Network Service Provider may not recover charges under this clause to the extent these are:

(1) recovered through the Distribution Network Service Provider's annual revenue requirement;

(2) recovered under modified clauses 6.18.7A(a) to (c) or clause 6.18.7B; or

(3) recovered from another Distribution Network Service Provider.

2.1.6 Recovery of charges for Climate Change Fund payments

In accordance with clause 6.18.7B of the *Transitional Rules*, Endeavour Energy's 2012/13 Annual Pricing Proposal must satisfy the following criteria:

6.18.7B Recovery of Climate Change Fund payments

(a) In the case of a NSW Distribution Network Service Provider, a pricing proposal must provide for tariffs designed to pass on to customers Climate Change Fund payments.

(b) The amount to be passed on to customers for a particular regulatory year must not exceed the estimated amount of Climate Change Fund payments for the relevant regulatory year adjusted for over or under recovery in any previous regulatory year.

(c) The extent of the over or under recovery is the difference between:

(1) the actual amount of the Climate Change Fund payments in a previous regulatory year; and

(2) the amount passed on to customers in respect of Climate Change Fund payments in that previous regulatory year.

2.2 Requirements of the AER's 2009 Distribution Determination

2.2.1 Weighted Average Price Cap

In accordance with section 4.6 of the *2009 Distribution Determination*, Endeavour Energy must submit to the AER proposed tariffs and charging parameters which correspond to the price terms contained in the Weighted Average Price Cap (WAPC) equation as set out on the following page.

$$\frac{\sum_{i=1}^n \sum_{k=1}^m p_{ik}^t \times q_{ik}^{t-2}}{\sum_{i=1}^n \sum_{k=1}^m p_{ik}^{t-1} \times q_{ik}^{t-2}} \leq (1 + \Delta CPI_t) \times (1 - X_t) \times (1 + D_t) \pm (\text{pass through}_t) \quad i = 1, \dots, n \text{ and } k = 1, \dots, m.$$

Where: The DNSP has 'n' relevant tariff classes which each have up to 'm' components:

p_{ik}^t is the proposed price for component 'k' of the relevant tariff 'i' for year 't'

p_{ik}^{t-1} is the actual price for component 'k' of the relevant tariff 'i' for year 't-1' (being the year which immediately precedes year 't')

q_{ik}^{t-2} is the audited¹⁹⁹ quantity of component 'k' of the relevant tariff 'i' that was charged by the DNSP in year 't-2' (being the year immediately preceding year 't-1')

X_t is the allowed real change in average prices from year 't-1' to year 't' of the regulatory control period as determined by the AER

D_t is the demand management cost recovery factor for year 't' calculated to recover certain approved demand management implementation costs and foregone revenue incurred in year 't-2',²⁰⁰

Pass through_t represents approved pass through amounts (expressed in percentage form) with respect to regulatory year 't' as determined by the AER under clause 6.6 of the transitional chapter 6 rules and chapter 15 of this final decision.

ΔCPI_t means the number derived, with respect to regulatory year 't', from the application of the following formula:

$$\Delta CPI_t = \frac{CPI_{March(t-2)} + CPI_{June(t-2)} + CPI_{September(t-1)} + CPI_{December(t-1)}}{CPI_{March(t-3)} + CPI_{June(t-3)} + CPI_{September(t-2)} + CPI_{December(t-2)}} - 1$$

2.2.2 Side Constraint

In accordance with section 4.6 of the *2009 Distribution Determination*, Endeavour Energy must submit to the AER proposed tariffs and charging parameters which correspond to the price terms contained in the side constraint equation as set out on the following page.

$$\frac{\sum_{k=1}^m d_k^t \times q_k^{t-2}}{\sum_{k=1}^m d_k^{t-1} \times q_k^{t-2}} \leq (1 + \Delta CPI_t) \times (1 - X_t) \times (1 + D_t) \times (1 + 2\%) \pm (\text{pass through}_t) \quad k = 1, \dots, m.$$

Where: The tariff class has up to 'm' components:

d_k^t is the proposed price for component 'k' of the tariff class for year 't'

d_k^{t-1} is the price charged by the DNSP for component 'k' of the tariff in year 't-1'

q_k^{t-2} is the audited quantity of component 'k' of the tariff that was charged by the DNSP in year 't-2'

X_t is the allowed real change in average prices from year 't-1' to year 't' of the regulatory control period as determined by the AER. If $X > 0$, then X will be set equal to zero for the purposes of the side constraint formula.

D_t is the demand management cost recovery factor for year 't' calculated to recover certain approved demand management implementation costs and foregone revenue incurred in year 't-2',²⁰¹

Pass through_t represents approved pass through amounts (expressed in percentage form) with respect to regulatory year 't' as determined by the AER under rule 6.6 of the transitional chapter 6 rules and chapter 15 of this final decision.

ΔCPI_t means the number derived, with respect to regulatory year 't', from the application of the following formula:

$$\Delta CPI_t = \frac{CPI_{March(t-2)} + CPI_{June(t-2)} + CPI_{September(t-1)} + CPI_{December(t-1)}}{CPI_{March(t-3)} + CPI_{June(t-3)} + CPI_{September(t-2)} + CPI_{December(t-2)}} - 1$$

In accordance with the AER's guideline on control mechanisms for direct control services¹, the side constraints have been applied starting from the second regulatory year of the regulatory control period (2010/11). Therefore, the tariff class side constraint is applicable to Endeavour Energy's tariff classes for 2012/13 and has been addressed in this pricing proposal.

¹ AER (2008) Final Decision on Control Mechanisms for Direct Control Services for the ACT and NSW 2009 Distribution Determinations, pg 12

2.2.3 Revenue requirement and X factors

The following table outlines the AER's final determination on Endeavour Energy's revenue requirements and pricing X factors for the *2009 regulatory control period*.²

Table 2.1 – AER conclusion on annual revenue requirements and X factors (\$m, nominal)

	2008/09	2009/10	2010/11	2011/12	2012/13	2013/14
Regulatory Depreciation		144.3	123.2	119.8	113.5	106.3
Return on Capital		369.8	415.9	470.0	518.5	563.5
Tax Allowance		38.8	42.9	43.1	42.9	43.5
Operating Expenditure		304.8	314.8	327.4	339.7	346.8
Annual Revenue Requirements		857.7	896.8	960.4	1,014.6	1,060.1
Smoothed Revenues	652.8	749.9	874.3	1,023.9	1,077.6	1,101.7
Forecast CPI (%)		2.47	2.47	2.47	2.47	2.47
X Factors ^a (%)		-12.58	-13.00	-13.00	-0.15	1.72

(a) Negative values for X indicate real price increases under the CPI-X formula.

² AER, *2009 distribution determination*, Table 16.27 updated for Australian Competition Tribunal Merits Review decision, 2009. Application by EnergyAustralia and Others (No 2) [2009] ACompT9.

BUSINESS CHARACTERISTICS

3.1 Summary

Endeavour Energy is a NSW based state owned energy corporation, serving some of Australia's largest and fastest growing regional economies. It provides distribution network services to just over 871,000 customers, or 2.2 million people, in households and businesses across a network franchise spanning 24,500 square kilometres in Greater Western Sydney, the Blue Mountains, the Illawarra and the Southern Highlands.

Endeavour Energy provides distribution network services to approximately 792,000 residential customers and 75,000 small business customers. There are also 4,750 large customers, with annual electricity consumption in excess of 160 MWh per annum, located within the Endeavour Energy network supply area.

In effectively managing distribution network services to its customers, Endeavour Energy takes account of a range of regional, climatic, customer, asset and licence requirement issues. Some of the key issues affecting Endeavour Energy's business include:

- The region in which Endeavour Energy operates includes some of Australia's fastest growing communities, in terms of population and growth. Population in the region is forecast to grow by 6% over the 2009/10 to 2013/14 determination period, while maximum demand for electricity is forecast to increase by 15% over the same period;
- Land use within the Endeavour Energy network supply region has also been steadily shifting from rural and semi-rural to urban and light commercial. As a result, customers now expect improved reliability performance in these areas;
- Peak temperatures within the Endeavour Energy network supply region are typically higher and more sustained than those of central Sydney and other coastal areas and, crucially, are more consistently experienced on an annual basis, meaning extreme weather events are more likely;
- Combined with these factors, the way Endeavour Energy's customers are using electricity has shifted rapidly over the last ten years. 81%³ of households in the Greater Western Sydney region are now fitted with air conditioning units,

³ Based on customer survey analysis - iView Telephone Survey, June 2008.

compared with 51% in the Illawarra and 50% in the Blue Mountains. This use has resulted in a peakier load pattern for Endeavour Energy in hot weather requiring more assets to service demand that only exists for a very small time frame each year;

- In addition to the increase in the penetration of air conditioning units, the uptake of other appliances such as in-home computers and large screen televisions has also increased. This trend is also changing consumption patterns, resulting in heightened customer expectations and awareness about reliability and security of supply;
- The impact of the continuing high growth rates and changing electricity patterns has meant deterioration in the overall system load factor; and
- From an asset perspective, in addition to operating in more extreme conditions, many of Endeavour Energy's assets are now nearing the end of their effective lives.

3.2 Endeavour Energy's business

Endeavour Energy is a state owned energy corporation in NSW, incorporated under the Energy Services Corporations Act 1995 and currently operates as a licensed distribution network service provider under the terms of the Electricity Supply Act 1995.

Endeavour Energy's network is one of the largest in Australia and serves just over 871,000 customers, or 2.2 million people, across 24,500 square kilometres.

The network currently comprises 30,181 transmission, zone and distribution substations and 307,000 power poles connected by 47,000 kilometres of underground and overhead lines.

Figure 3.1: Endeavour Energy's network



3.3 Characteristics of the region

Endeavour Energy's network covers the Greater Western Sydney, Blue Mountains, Illawarra and Southern Highlands areas. It includes the major cities of Parramatta, Penrith, Wollongong, Campbelltown, Blacktown and Liverpool and spans 18 local government areas.⁴

3.3.1 Regional growth and changing land use

Endeavour Energy supplies power to some of Australia's fastest growing communities, in terms of both population and load. Land use has been steadily moving from rural and semi-rural to generally urban and light commercial (but importantly not heavy industrial).

⁴ Lithgow, Hawkesbury, Blue Mountains, Penrith, Baulkham Hills, Blacktown, Parramatta, Fairfield, Holroyd, Liverpool, Camden, Campbelltown, Wollondilly, Wingecarribee, Wollongong, Shellharbour, Kiama, Shoalhaven

This trend is continuing with new release areas in the north-west, including North Kellyville and Second Ponds Creek and new releases at the ADI site, Employment Lands at Eastern Creek and the south-west sector, such as Oran Park and Turner Road. The following aerial photos of Kellyville in Sydney's north-west between 1996 and 2005 are representative of the significant effect on Endeavour Energy's network required to service this change in land use.

Figure 3.2: Increasing urbanisation of region

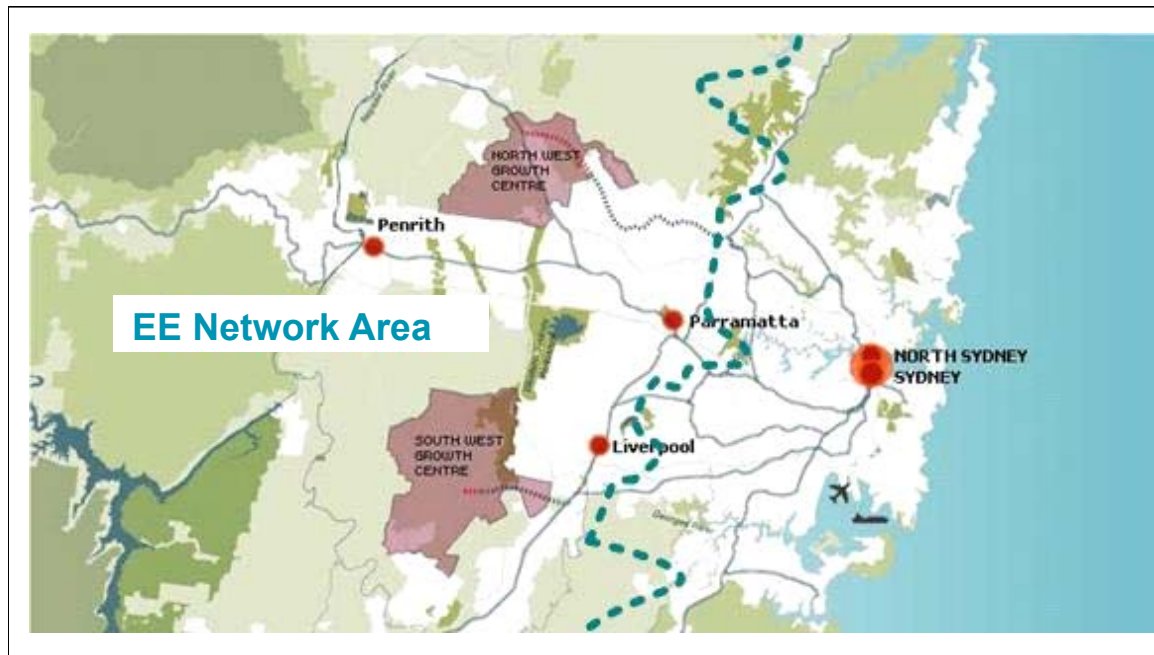


Major land releases are also occurring at Dapto and Shellcove in the Illawarra region. Combined with ongoing urban consolidation, these trends create pressure to raise network reliability performance from levels previously accepted as appropriate for rural areas, to levels now expected by customers in more densely populated urban areas.

The South West and North West Growth Centres are a key plank of the NSW Government's Metropolitan Strategy for Sydney. The Growth Centres will accommodate 30% to 40% of Sydney's long term housing growth and will eventually accommodate 181,000 new dwellings. Both the South West and North West Growth Centres are encompassed wholly within Endeavour Energy's network franchise. Figure 3.3 highlights the scale and location of a number of the major land releases that fall within Endeavour Energy's franchise area.

These changes will have a significant impact on Endeavour Energy's network over the longer term. It is anticipated that the population serviced will grow by approximately 6% over the 2009/10 to 2013/14 determination period, increasing maximum demand by 15% over the same period and requiring sustained investment in the network.

Figure 3.3: Major developments in Endeavour Energy's network



3.3.2 Regional climatic considerations

The Endeavour Energy distribution area, particularly the growth corridors to the west of Sydney, experience peak temperatures that are typically higher and more sustained than those of central Sydney and other coastal areas, and crucially are more consistently experienced on an annual basis by the network. There is a particularly high probability of experiencing extreme temperature events (in excess of 40°C) in each year in the west of Sydney.

The design temperature⁵ for meeting peak demand on the network in a standard summer is 41°C for the western regions in Endeavour Energy's network, compared to only 30°C for Endeavour Energy's coastal regions.

Moreover, in Endeavour Energy's distribution area, the extreme rating temperature for the system⁶ (similar to a 1 in 10 year type event) is only marginally higher at 43°C, so that there is relatively very little difference (just 2°C) between a standard summer peak temperature and an extreme summer peak temperature. This means there is a very high probability (almost certainty given design constraints) that there will be an extreme temperature experienced in any given year in the Endeavour Energy network. Endeavour Energy needs to invest in its network to ensure acceptable service standards under these extremes of load and temperature.

⁵ 50% Probability of Exceedance (PoE) – one in two year event.

⁶ Effectively the 10% probability of exceedance temperature.

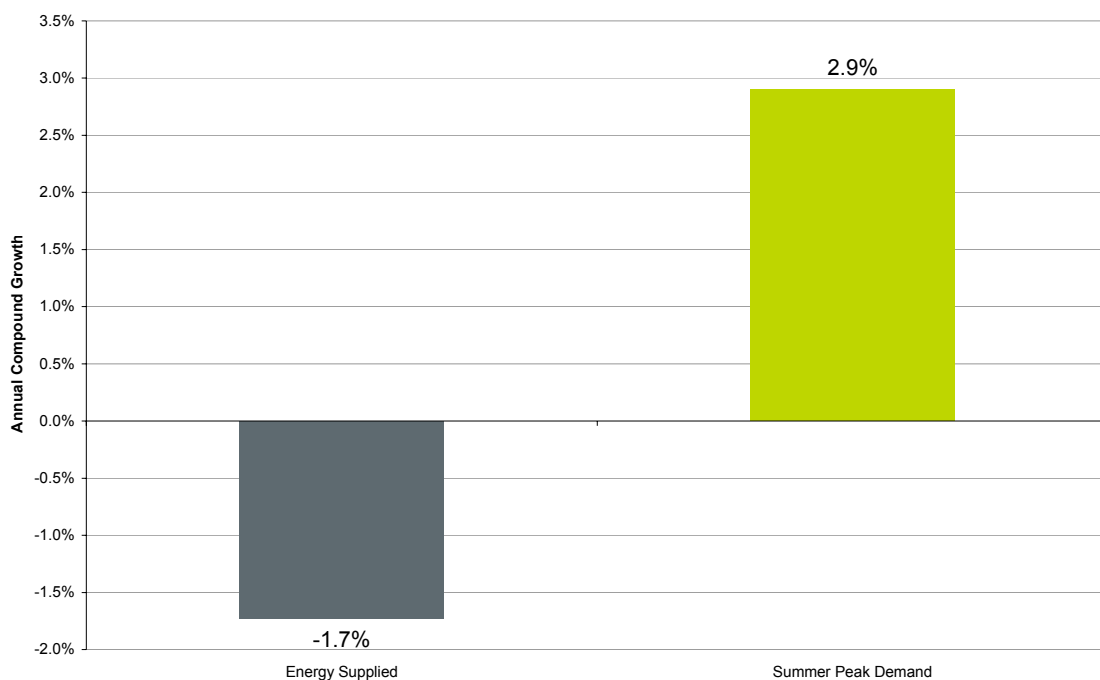
3.4 Endeavour Energy's customer and demand profile

Endeavour Energy serves around 792,000 residential customers, 75,000 small business customers and 4,750 large customers with annual electricity consumption in excess of 160 MWh pa.

3.4.1 Customer usage trends and the predominance of air conditioning

The way customers use electricity significantly affects the development of the electricity network. The higher and more sustained temperatures experienced in the west of Sydney primarily impact the peak demand growth rate for Endeavour Energy. While demand continues to grow, energy consumption over the period 2007/08 to 2011/12 has been in decline, as highlighted in Figure 3.4.

**Figure 3.4: Energy and demand growth rates
(2007/08 to 2011/12)**



The increased uptake of air conditioning in new residential developments and the increasing penetration of air conditioning units into existing dwellings, coupled with an increasing air conditioning (kW) loading as customers become more accustomed to using refrigerated air in their homes, are all causing demand to increase at a faster rate than energy growth. This trend is expected to continue through the *2009 regulatory control period* before it ultimately plateaus with appliance saturation.

Figure 3.5 shows the effect of the air conditioning load on peak demand on an extreme temperature day relative to a mild summer day. The increase in load on an extreme temperature day is largely due to domestic and general supply non time of use customers.

Figure 3.5: Peak demand on an extreme temperature day relative to mild summer day



Even though these peak demands occur relatively infrequently, Endeavour Energy is required to ensure that the network is capable of meeting this demand when it does occur. This means that some network assets are used only to service the air conditioning demand of households during the extreme weather events and only for very short periods e.g. typically less than 85 hours per year.

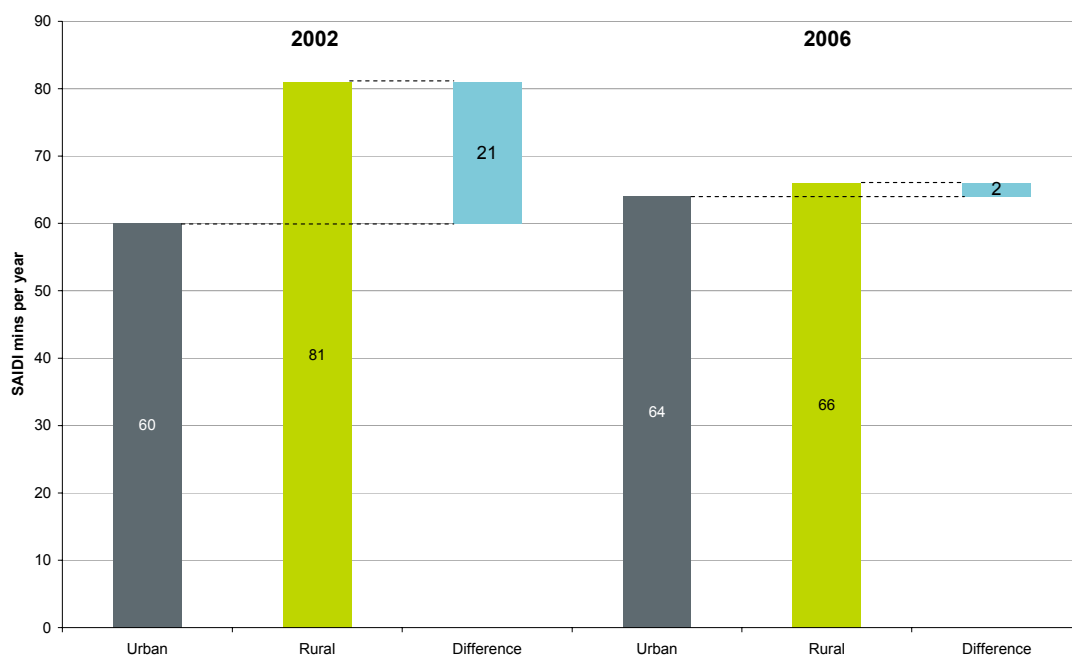
3.4.2 Customer service expectations

Customer expectations about network reliability are increasing. Endeavour Energy commissioned research studies into customer expectations of reliability in 2002 and again in 2006. One study involved choice modelling to understand the level of electricity reliability that customers were willing to trade off against other offerings.

The research shows that while urban customers' expectations have remained steady, rural customers' expectations about the number and duration of outages have changed significantly to a point where they are now very similar to those of urban customers. This trend in customer expectations is shown in Figure 3.6 and is consistent with the increasing use of home computers and other technology.

Endeavour Energy recognises the increased expectations of rural customers. The NSW DRP Licence Conditions establish interruption duration and frequency standards that apply to Endeavour Energy feeders and customers. Endeavour Energy has established a program to ensure that rural feeders are operated to comply with these conditions.

Figure 3.6: Urban and rural customer expectations – SAIDI⁷ minutes per year without supply (average)



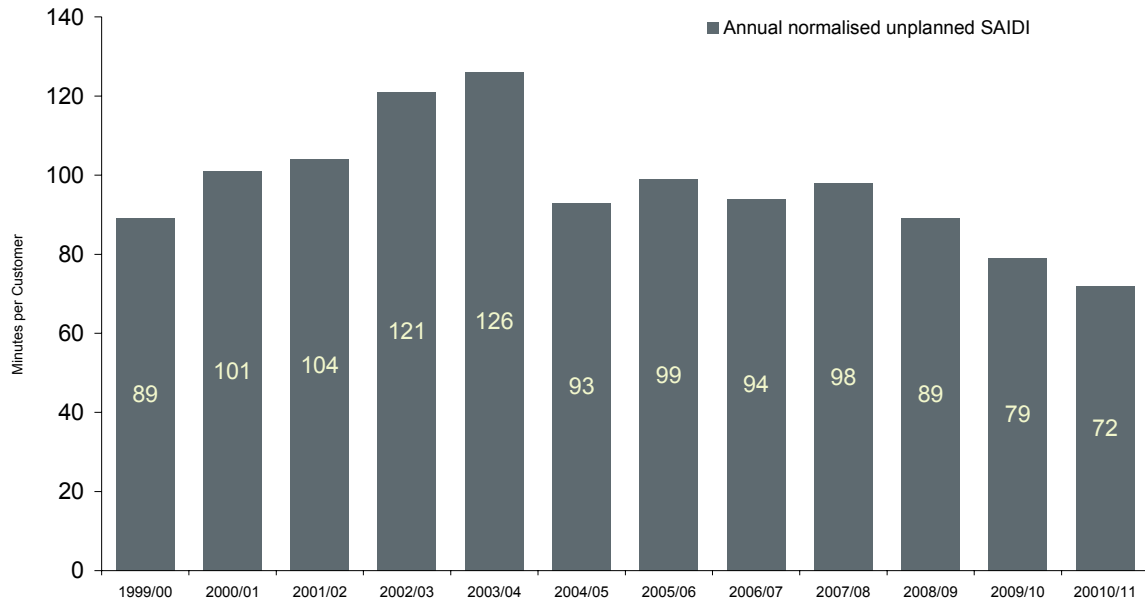
3.4.3 Network reliability

Improving reliability performance has been one of the key drivers of expenditure for Endeavour Energy. The total system average interruption duration index (SAIDI) measures the time, in minutes, that an average customer is without power in a given year. This is an important measure of an electricity distribution network's performance.

Over the period 1999/00 to 2003/04, Endeavour Energy customers had experienced declining performance, reaching a SAIDI of 126 minutes in 2003/04, after which reliability improved by almost 43% to 72 minutes in 2010/11. The improvement in reliability has been achieved through a combination of targeted maintenance and capital initiatives. Endeavour Energy's 2004 regulatory submission set out plans to continue to focus on improving performance in its worst performing areas (continuing a strategy that commenced in 2002), by managing load at risk and planning for and replacing aged network assets. These plans were implemented and network reliability improved as shown in Figure 3.7.

⁷ System Average Interruption Duration Index

Figure 3.7: Network reliability performance



These characteristics serve to highlight some of the key factors that Endeavour Energy has considered in developing its Network Tariff Strategy. Endeavour Energy's Network Tariff Strategy is underpinned by its network planning and expenditure forecasting processes and articulates the future priorities for the network business. The process for developing the Network Tariff Strategy and ensuring that Endeavour Energy is appropriately prepared to respond to these issues is discussed further in Chapter 5.

4

TARIFF CLASSES

4.1 Regulatory requirements

In accordance with clause 6.18.3 of the *Transitional Rules*, Endeavour Energy must satisfy the following criteria with respect to tariff classes:

6.18.3 Tariff classes

- (a) A pricing proposal must define the tariff classes into which customers for direct control services are divided.*
- (b) Each customer for direct control services must be a member of 1 or more tariff classes.*
- (c) Separate tariff classes must be constituted for customers to whom standard control services are supplied and customers to whom alternative control services are supplied (but a customer for both standard control services and alternative control services may be a member of 2 or more tariff classes).*
- (d) A tariff class must be constituted with regard to:*
 - (1) the need to group customers together on an economically efficient basis; and*
 - (2) the need to avoid unnecessary transaction costs.*

4.2 Summary of Tariff Classes

Endeavour Energy's network use of system tariffs represent an aggregation of distribution use of system tariffs, climate change fund recovery tariffs and transmission cost recovery tariffs. Endeavour Energy's tariff classes are applicable to all of the above mentioned tariffs.

The following table provides a summary of Endeavour Energy's standard control services tariff classes⁸.

⁸ Endeavour Energy's Alternate Control Services are discussed in Chapter 13 of this pricing proposal

Table 4.1 – Endeavour Energy’s network tariff classes

Tariff Class	Pricing Options	Connection Characteristics
Low Voltage Energy	Residential Inclining Block Tariff	LV Connection; <160MWh pa
	Residential Time of Use	LV Connection; <160MWh pa
	General Supply Inclining Block Tariff	LV Connection; <160MWh pa
	General Supply Time of Use	LV Connection; <160MWh pa
	Controlled Load 1	LV Connection; <160MWh pa
	Controlled Load 2	LV Connection; <160MWh pa
Low Voltage Demand	LV TOU Demand	LV Connection; >160MWh pa but <40GWh or 10MVA
	LV TOU Demand Transition Tariff	LV Connection; >160MWh pa but <40GWh or 10MVA
High Voltage Demand	HV TOU Demand	HV Connection; >160MWh pa but <40GWh or 10MVA
	Individually Calculated	HV Connection; >40GWh or 10MVA
Subtransmission Demand	ST TOU Demand	ST Connection; >160MWh pa but <40GWh or 10MVA
	Individually Calculated	ST Connection; >40GWh or 10MVA
Inter-Distributor Transfers Demand	Individually Calculated	Distributor Transfer
Unmetered Supply	Streetlighting	Unmetered
	Traffic Control Signals (New in 2012/13)	Unmetered
	Nightwatch	Unmetered

Endeavour Energy has defined tariff classes with respect to:

- The nature and extent of customers’ usage;
- The nature of the customers’ connection to the network; and
- Whether remotely-read interval metering or other similar metering technology has been installed at the customer’s premises as a result of a regulatory obligation or requirement.

Endeavour Energy believes that this definition of tariff classes represents an economically efficient tariff class assignment that will avoid the need for unnecessary transaction costs.

These tariff classes are consistent with those included in Endeavour Energy’s 2011/12 Annual Pricing Proposal, as approved by the AER on 3 June 2011.

4.3 Low Voltage Energy Tariff Class

4.3.1 Residential inclining block tariff – N70

The residential network use of system inclining block tariff (IBT) applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is less than 160MWh; and
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V.

In addition, the residential IBT is predominantly used for one or more of the following purposes:

- Private dwellings;
- Boarding and lodging houses, being any house in which three or more persons exclusive of the family of the proprietor thereof are lodged for hire or reward from week to week or for more than a week;
- Retirement villages;
- Residential sections of nursing homes and hospitals;
- Residential sections of educational institutions;
- Children's homes;
- Approved baby health centres, day nurseries and kindergartens;
- Churches, mosques, temples etc., being buildings or properties which are used principally for public worship or partly for public worship and partly for educational purpose; or
- Approved caravan sites.

The residential IBT is applied where the *connection point* has an accumulation (basic or disc - Type 6) meter or an interval meter that is read as a Type 6 meter.

The residential IBT comprises the following components:

- A Network Access Charge (\$pa); and
- Two variable energy components consisting of:
 - a) a First Block Rate, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) up to and including 1,750 kWh per quarter; and
 - b) a Second Block Rate, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) above 1,750 kWh per quarter.

4.3.2 Residential time of use (type 5) tariff – N705

The residential network use of system time of use (type 5) tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is less than 160MWh;
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V;
- A capital contribution for the meter is payable by the customer; and
- The interval meter records 30 minute intervals.

The residential time of use (type 5) tariff applies to a property, which is predominantly used for one or more of the purposes set out in section 4.3.1 at a *connection point* with a time of use meter from which interval meter data is obtained. Type 5 tariffs are applicable to supply points with a Type 5 (manually read interval) meter installed.

The residential time of use (type 5) tariff comprises the following components:

- A Network Access Charge (\$pa); and
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.

The time of day period definitions are outlined in Appendix I.

4.3.3 Residential time of use tariff – N706

The residential network use of system time of use tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is less than 160MWh;
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V; and
- The interval meter records a single “peak”, “shoulder” and “off peak” interval per billing period.

The residential time of use tariff applies to a property, which is predominantly used for one or more of the purposes set out in section 4.3.1 at a *connection point* with a time of use meter from which interval meter data is obtained.

The residential time of use tariff comprises the following components:

- A Network Access Charge (\$pa); and
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.

The distinctions between this TOU tariff (N706) and the Type 5 TOU Tariff (N705) discussed above are that:

- Only three periods (one each for Peak, Shoulder and Off-Peak) are recorded for each meter reading cycle under tariff N706, rather than 48 periods per day (i.e. 30 minute intervals) that are collected for the Type 5 TOU Tariff (N705); and
- A capital contribution is payable for the meter under the Type 5 TOU Tariff (N705).

The time of day period definitions are outlined in Appendix I.

4.3.4 General supply inclining block tariff – N90

The general supply network use of system inclining block tariff (IBT) applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is less than 160MWh;
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V.

The general supply IBT applies to low voltage electricity used for any purpose other than Residential, at a *connection point* with an accumulation meter or an interval meter that is read as a Type 6 meter.

The general supply IBT comprises the following components:

- A Network Access Charge (\$pa); and
- Two variable energy components consisting of:
 - a) a First Block Rate, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) up to and including 2,500 kWh per quarter; and
 - b) a Second Block Rate, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) above 2,500 kWh per quarter.

4.3.5 General supply time of use (type 5) – N845

The general supply network use of system time of use (type 5) tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is less than 160MWh;
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V;
- A capital contribution for the meter is payable by the customer; and
- The interval meter records 30 minute intervals.

The general supply time of use (type 5) tariff applies to a property, which is predominantly used for any purpose other than residential at a *connection point* with a time of use meter from which interval meter data is obtained. Type 5 tariffs are applicable to supply points with a Type 5 (manually read interval) meter installed.

The general supply time of use (type 5) tariff comprises the following components:

- A Network Access Charge (\$pa); and
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.

The time of day period definitions are outlined in Appendix I.

4.3.6 General supply time of use – N84

The general supply network use of system time of use tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is less than 160MWh;
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V; and
- The interval meter records a single “peak”, “shoulder” and “off peak” interval per billing period.

The general supply time of use tariff applies to a property, which is predominantly used for any purpose other than residential at a *connection point* with a time of use meter from which interval meter data is obtained.

The general supply time of use tariff comprises the following components:

- A Network Access Charge (\$pa); and
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.

The distinctions between this General Supply TOU tariff (N84) and the General Supply Type 5 TOU Tariff (N845) discussed above are that:

- Only three periods (one each for Peak, Shoulder and Off-Peak) are recorded for each meter reading cycle under tariff N84, rather than 48 periods per day (i.e. 30 minute intervals) that are collected for the Type 5 TOU Tariff (N845); and
- A capital contribution is payable for the meter under the Type 5 TOU Tariff (N705).

The time of day period definitions are outlined in Appendix I.

4.3.7 Controlled load tariffs – N50 and N54

A control load network use of system tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is less than 160MWh;
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V; and
- A Residential or general supply tariff also applies.

A controlled load tariff applies where electricity load is separately metered and controlled at a *connection point* with an accumulation meter.

- a) Controlled Load 1 applies where specified appliances are controlled by our equipment so that supply may not be available between 7:00am and 10:00pm; and
- b) Controlled Load 2 applies where our equipment controls approved specified appliances so that electricity is available for restricted periods not exceeding a total of 17 hours in any period of 24 hours.

Switching times will be managed to minimise network investment and meet customer needs for the load being controlled.

Customers with a Controlled Load relay are entitled to a controlled load network price only if all of the following conditions are met:

- a) Controlled Load consumption is separately metered using the same type of meter as the uncontrolled portion of a customer's load;
- b) Controlled Load consumption and uncontrolled load consumption is always synchronously read, i.e. on the same day; and
- c) The Controlled Load is operated by our load control systems or time switches.

A controlled load tariff is applicable only to approved appliances as detailed in Appendix 2 of Endeavour Energy's Network Price List.

The controlled load 1 and 2 tariffs comprise of the following components:

- A Network Access Charge (\$pa); and
- An Energy rate, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh).

4.4 Low Voltage Demand Tariff Class

4.4.1 Low voltage time of use demand – N19

The low voltage network use of system time of use demand tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is greater than 160MWh but less than 40 GWh or 10MVA;
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V; and
- There exists a time of use meter, from which both interval meter energy and demand data is obtained.

The low voltage time of use demand tariff comprises the following components:

- A Network Access Charge (\$pa);
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.
- Two variable demand components consisting of:

- a) a High-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the High-Season Peak period; and
- b) a Low-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the Low-Season Peak period.

The time of day and season definitions are outlined in Appendix I.

4.4.2 Transitional time of use – N89

The transitional network use of system time of use tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, greater than 160MWh but less than 40 GWh or 10MVA;
- Electricity is supplied at a voltage level defined as Low Voltage (LV) - nominally 230/400 V; and
- There exists a time of use meter, from which at a minimum interval meter energy data is obtained.

The Transitional time of use tariff applies to those customers on a Low Voltage Energy Tariff Class tariff who have been identified as consuming in excess of 160MWh per annum, and therefore requiring a demand based tariff, but cannot be directly transferred to the low voltage time of use demand tariff due to a lack of metering capable of supporting the demand based tariff or the expected bill impact of a direct transition to low voltage time of use demand is deemed excessive.

The transitional time of use tariff comprises the following components:

- A Network Access Charge (\$pa); and
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.

The time of day period definitions are outlined in Appendix I.

4.5 High Voltage Demand Tariff Class

4.5.1 High voltage time of use demand – N29

The high voltage network use of system time of use demand tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, greater than 160MWh but less than 40 GWh or 10MVA; and
- Electricity is supplied at a voltage level defined as High Voltage (HV) - nominally 12.7 kV SWER, 11 or 22 kV.

The high voltage time of use demand tariff comprises the following components:

- A Network Access Charge (\$pa);
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.
- Two variable demand components consisting of:
 - a) a High-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the High-Season Peak period; and
 - b) a Low-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the Low-Season Peak period.

The time of day and season definitions are outlined in Appendix I.

4.5.2 Individually calculated high voltage time of use demand tariffs

Individually calculated high voltage network use of system time of use demand tariffs apply to *connection services* supplied to the *connection point* where:

- Electricity is supplied at a voltage level defined as High Voltage (HV) - nominally 12.7 kV SWER, 11 or 22 kV; and
- Electricity consumption has been equal to or greater than 100 GWh in total for the 36 months preceding the application; or
- Electricity consumption has been equal to or greater than 40 GWh per annum in each of the two financial years preceding the application; or
- Monthly peak demand has been equal to or greater than 10 MVA for 24 of the 36 months preceding the application.

Endeavour Energy may assign an individually calculated high voltage time of use demand tariff to any *connection point* in circumstances such as, but not limited to:

- The need to recover investment associated with stranded or dedicated assets, or other cost caused to Endeavour Energy by the customer at that *connection point*, which may otherwise not be recovered under the high voltage time of use demand tariff; and
- Endeavour Energy agrees to assign an individually calculated high voltage time of use demand tariff following an application from the customer, or a retail supplier acting on their behalf, to be assigned to an individually calculated high voltage time of use demand tariff.

Individually calculated high voltage time of use demand tariffs comprise the following components:

- A Network Access Charge (\$pa);
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.
- Two variable demand components consisting of:
 - a) a High-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the High-Season Peak period; and
 - b) a Low-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the Low-Season Peak period.

The time of day and season definitions are outlined in Appendix I.

4.6 Subtransmission Voltage Demand Tariff Class

4.6.1 Subtransmission time of use demand – N39

The subtransmission network use of system time of use demand tariff applies to *connection services* supplied to the *connection point* where:

- Total electricity consumption, per financial year, is greater than 160MWh but less than 40 GWh or 10MVA; and
- Electricity is supplied at a voltage level defined as Sub-Transmission (ST) - 33, 66 or 132 kV.

The subtransmission time of use demand tariff comprises the following components:

- A Network Access Charge (\$pa);

- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.
- Two variable demand components consisting of:
 - a) a High-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the High-Season Peak period; and
 - b) a Low-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the Low-Season Peak period.

The time of day and season definitions are outlined in Appendix I.

4.6.2 Individually calculated subtransmission voltage tariffs

Individually calculated subtransmission network use of system time of use demand tariffs apply to *connection services* supplied to the *connection point* where:

- Electricity is supplied at a voltage level defined as Sub-Transmission (ST) - 33, 66 or 132 kV; and
- Electricity consumption has been equal to or greater than 100 GWh in total for the 36 months preceding the application; or
- Electricity consumption has been equal to or greater than 40 GWh per annum in each of the two financial years preceding the application; or
- Monthly peak demand has been equal to or greater than 10 MVA for 24 of the 36 months preceding the application.

Endeavour Energy may assign an individually calculated subtransmission time of use demand tariff to any *connection point* in circumstances such as, but not limited to:

- The need to recover investment associated with stranded or dedicated assets, or other cost caused to Endeavour Energy by the customer at that *connection point*, which may otherwise not be recovered under the subtransmission time of use demand tariff; and
- Endeavour Energy agrees to assign an individually calculated subtransmission time of use demand tariff following an application from the customer, or a retail supplier acting on their behalf, to be assigned to an individually calculated high voltage time of use demand tariff.

Individually calculated subtransmission time of use demand tariffs comprise the following components:

- A Network Access Charge (\$pa);
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.
- Two variable demand components consisting of:
 - a) a High-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the High-Season Peak period; and
 - b) a Low-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the Low-Season Peak period.

The time of day and season definitions are outlined in Appendix I.

4.7 Inter-Distributor Transfer Tariff Class

Inter-distributor transfer network use of system tariffs are calculated on an individual basis and are specifically applied to electricity transferred through the Endeavour Energy network on behalf of Ausgrid and Essential Energy.

Inter-distributor transfer tariffs comprise the following components:

- A Network Access Charge (\$pa);
- Three variable energy components consisting of:
 - a) a Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Peak period;
 - b) a Shoulder Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Shoulder period; and
 - c) an Off-Peak Energy rate, expressed on a ¢/kWh basis, to be applied to the consumption of electricity in the Off-Peak period.
- Two variable demand components consisting of:
 - a) a High-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the High-Season Peak period; and

- b) a Low-Season Peak Demand rate, expressed on a \$/maximum kVA/month basis, applied to the consumption of electricity in the Low-Season Peak period.

The time of day and season definitions are outlined in Appendix I.

4.8 Unmetered Supply Tariff Class

Endeavour Energy offers an unmetered streetlighting supply tariff for streetlighting *connection points* that are not metered.

Energy consumption is calculated using the appropriate algorithm in the applicable Metrology Procedure.

The streetlighting tariff comprises an Energy rate, expressed on a ¢/kWh basis, to be applied to calculated electricity consumption (kWh).

As part of this *2012/13 Annual Pricing Proposal*, Endeavour Energy has proposed the introduction of a new unmetered traffic control signal light tariff for traffic control signal light *connection points* that are not metered. This tariff will comprise an Energy rate, expressed on a ¢/kWh basis, to be applied to calculated electricity consumption (kWh).

4.9 Miscellaneous, monopoly service and emergency recoverable works charges

4.9.1 Miscellaneous service charges

In accordance with Appendix H of the *2009 Distribution Determination*, Distribution Network Service Providers can levy various charges for services defined as miscellaneous services.

Endeavour Energy's miscellaneous service charges are provided in Chapter 6 of this *2012/13 Annual Pricing Proposal*.

4.9.2 Monopoly service charges

In accordance with Appendix H of the *2009 Distribution Determination*, Distribution Network Service Providers can levy various charges for services defined as monopoly services.

Endeavour Energy's monopoly service charges are provided in Chapter 6 of this *2012/13 Annual Pricing Proposal*.

4.9.3 Emergency recoverable works

In accordance with Appendix H of the *2009 Distribution Determination*, Distribution Network Service Providers can levy various charges for services defined as emergency recoverable works.

Endeavour Energy's charges for emergency recoverable works charges are provided in Chapter 6 of this *2012/13 Annual Pricing Proposal*.

5

NETWORK TARIFF STRATEGY

5.1 Network Tariff Objectives

To provide a high-level framework for network tariff strategy development the following network pricing objectives have been adopted:

Table 5.1– Network Pricing Objectives

Network Pricing Objective	Driver
Revenue sufficiency	Fund our investment programs and ensure that Endeavour Energy recovers the regulated revenue allowance, set to reflect efficient costs.
Revenue stability	Actively seek to minimise monthly and annual revenue variations where practical.
Pricing efficiency	Signal the economic cost of network peak demand and utilisation and recover fixed/sunk costs in a manner that causes the least distortion to price signals.
Equitable customer outcomes	Ensure customers only pay for their fair share of costs and that any moves to cost reflectivity are introduced in a socially responsible manner.

5.2 The Need for Tariff Reform

Air conditioning demand is highly sensitive to peak summer temperatures, has very poor power factors and occurs when the system is significantly de-rated because of high summer temperatures. Increased air-conditioning load and its impact on peak demand is one of the key drivers of Endeavour Energy's network investment.

Capacity in Endeavour Energy's network must be built to meet maximum peak load unless demand can be managed or reduced. Capacity requirements for new estates have effectively doubled as a result of air conditioning load on peak demand – network design parameters, for example, have changed from 3.5 to 4.0 kVA to 6.0 to 7.5 kVA per household.

New houses are estimated to have approximately 60% to 70% penetration of large ducted air conditioning. Similarly there is a high take-up of air conditioning occurring in some existing areas as split systems in particular have become more affordable in recent years.

Modelling undertaken by Charles River Associates⁹ (CRA) for Endeavour Energy using a long run marginal cost approach for capital investment to meet peak demand indicates that the cross-subsidy between customers with air conditioning and those without air conditioning is in the range \$80 to \$110 million per year – or approximately one third of total sales to the residential and small business sectors.

If this subsidy is smeared across all of the remaining consumption by these groups it equates to 1.5 cents/kWh to 2.0 cents/kWh relative to the marginal rate.

5.3 Network Tariff Strategy

Endeavour Energy's pricing is designed in part to signal the costs associated with increased demand placed on the network, including the use of air conditioning, within metering and efficiency constraints, while recovering the necessary revenues to support our investment programs.

Consistent with the network tariff objectives outlined above, Endeavour Energy's network tariff strategy aims to:

- Attain revenue sufficiency under the Weighted Average Price Cap;
- Signal the long run marginal cost of supply through its network tariffs;
- Ensure tariffs are set in a socially responsible manner;
- Improve cost reflectivity by reducing the reliance on energy based tariff components where appropriate;
- Pass through the full cost of TransGrid's transmission services and preserve transmission price signals where possible; and
- Explore tariff based demand management opportunities, including voluntary time of use tariffs, and tariffs that target network constraints on a locational basis (see section 5.6 for future tariff reform options under consideration).

5.4 Tariff Reform Initiatives

Starting in 2004, Endeavour Energy has undertaken a number of tariff reforms, including the following:

- The introduction of the Inclining Block Tariff (IBT) Structure for residential and general supply customers;
- The introduction of a time of use tariff options for residential customers to complement the IBT;
- A new time of day (peak, shoulder and off-peak) period definition to better reflect current and expected patterns of network utilisation;
- The ongoing assignment of eligible customers to the low voltage time of use demand network tariff; and

⁹ CRA (2003), Impact of Air Conditioning on Endeavour Energy's Network.

- The introduction of seasonal peak demand charging to better reflect current and expected patterns of network utilisation.

5.5 Experimental Tariff Programs

Endeavour Energy's Western Sydney Pricing Trial commenced on 1 August 2006 and concluded on 31 July 2009. The trial involved approximately 1,300 residential customers from the Western Sydney region of Endeavour Energy's network. These customers were placed on either a Seasonal Peak Time of Use Tariff or Dynamic Peak Price Time of Use Tariff.

Those customers on the Dynamic Peak Pricing tariff were then broken into two treatment groups, one with an in home display unit and one without. A further group of customers on the default tariff was identified as a control group against which the tariff treatment groups could be compared.

The purpose of the trial was to test more sophisticated time of use pricing signals, in particular their:

- Effectiveness as a demand management tool;
- Acceptance by customers;
- Cost of administration; and
- Financial viability as a network-wide time of use tariff option.

The key results for the Dynamic Peak Time of Use Tariff (DPP) were:

- A demonstrable reduction in consumption and demand relative to the control group during the peak period 1-8pm on days when a dynamic peak event was called;
- A marginal conservation of energy for the entire period of the trial;
- No demonstrable response relative to the control group during shoulder and off peak periods;
- Customers do not appear to have shifted consumption from the peak to the off peak period;
- Customer response does not appear to vary during a dynamic peak event according to whether customers are given 2 hours or 24 hours notification of an impending event;
- Customer response does not appear to vary during a dynamic peak event if customers are subjected to continuous events called over several days; and
- Some evidence that providing customers on the dynamic peak tariff with an in-home display monitor will result in an additional 3% reduction in consumption during a dynamic peak event, however this diminished over time.

The results for the Seasonal Peak Time of Use Tariff suggest that a much reduced response rate is evident relative to the DPP trial.

Endeavour Energy will continue to undertake additional customer research as appropriate in order to move to more efficient prices in the medium term that signal the costs imposed on the network in times of peak demand.

5.6 Future Tariff Reform Options

Endeavour Energy believes that there may be significant benefits to be realised from network tariff reform and in obtaining more first-hand information on customers' usage of the network through trials and associated customer research. Endeavour Energy's key focus will be on the following network tariff reforms:

- Strengthening the IBT price signal as a "second best" option to TOU pricing where TOU meters are not available;
- Assigning more customers to network demand tariffs where it is economically efficient to do so;
- Exploring tariff based demand management opportunities; and
- Improving the design of Endeavour Energy's voluntary network time of use (TOU) tariffs and assessing the impacts on customers that have limited ability to change their usage patterns.

5.7 Expected DUOS Price Trends 2009 - 2014

Endeavour Energy's weighted average DUOS price is expected to increase over the *2009 regulatory control period* at the rate governed by the weighted average price cap X factors as determined by the AER in the *2009 Distribution Determination* and the *tariff class* side constraints as specified in the *Transitional Rules*.

In accordance with clause 6.18.9(a)(3) of the *Transitional Rules*, Endeavour Energy believes that in satisfying the regulatory requirements as demonstrated in Chapter 8 of this *2012/13 Annual Pricing Proposal* and implementing the above tariff strategy, customers can expect the DUOS *charging parameter* movements as provided in Table 5.2. The movements provided are relative to the weighted average price movement allowed by the AER in the *2009 Distribution Determination* over the *2009 regulatory control period* applied to all DUOS *charging parameter* movements.

It is also expected that the Residential IBT will increase at a lower rate than the General Supply IBT over the remainder of the *2009 regulatory control period*, within the constraints of the WAPC, applicable side constraints and avoidable and stand alone cost requirements.

Table 5.2 - Indication of future DUOS tariff charging parameter movements relative to the average price movement allowed by the AER in the 2009 Distribution Determination

DUOS Tariff	Fixed Charge	First Block Price	Second Block Price	Uniform Energy Price	Peak Energy Price	Shoulder Energy Price	Off-Peak Energy Price	High Season Maximum Demand	Low Season Maximum Demand
Residential IBT	▲	▲	▼						
Controlled Load 1	▲			▼					
Controlled Load 2	▲			▼					
Residential TOU	▲				▲	▼	▼		
General Supply IBT	▲	▲	▲						
General Supply TOU	▲				▲	▼	▲*		
LV TOU Demand	▲				▼	▼	▼	▲	▼
HV TOU Demand	▲				▼	▼	▼	▲	▼
ST TOU Demand	▲				▼	▼	▼	▲	▼

Legend:

▲ increase relative to the average overall distribution price movement allowed by the AER in the 2009 *distribution determination*

▼ decrease relative to the average overall distribution price movement allowed by the AER in the 2009 *distribution determination* (i.e. lower than the average distribution price increase, but may still result in a real increase at the component level)

— set at the average distribution price movement allowed by the AER in its 2009 *distribution determination*

* In the short term, the direction of change in this tariff component will be influenced by tariff alignment objectives.

5.8 Changes from the Previous Regulatory Year

In accordance with clause 6.18.2(b)(8) of the Transitional Rules, Endeavour Energy's 2012/13 Annual Pricing Proposal must describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination.

Endeavour Energy does not propose to make any changes to the structure of network tariffs in 2012/13 but does propose to introduce a new unmetered traffic control signal tariff from 1 July 2012. The new tariff will be applicable to traffic control signal lights owned and operated by the NSW Roads and Maritime Service.

The new tariff has been developed to address the unique nature and extent of the customer's usage following an extensive consultation process with representatives of the NSW Roads and Maritime Service.

5.9 Changes within the Regulatory Year

In accordance with clause 6.18.2(b)(5) of the Transitional Rules, Endeavour Energy's 2012/13 Annual Pricing Proposal must set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur.

Endeavour Energy does not propose to make any variations or adjustments to the structure of network tariffs during the course of 2012/13.

6

NETWORK USE OF SYSTEM TARIFFS

6.1 Calculation of Network Use of System Tariffs

Endeavour Energy's network use of system (NUOS) tariffs represent the aggregation of distribution use of system (DUOS) tariffs, climate change fund (CCF) recovery tariffs and transmission cost recovery (TCR) tariffs. The tariffs include the allowed movement in the consumer price index and are exclusive of GST.

6.2 Low Voltage Energy Tariff Class

6.2.1 Residential inclining block tariff – N70

The following table provides the proposed prices for the default residential inclining block tariff (IBT) for 2012/13. N70 is Endeavour Energy's primary residential tariff with over 99.9% of residential customers charged using this tariff.

Table 6.1 – Proposed 2012/13 residential inclining block tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	113.46	127.75	12.6%
First Block Energy Charge (c/kWh)	10.4280	10.7952	3.5%
Second Block Energy Charge (c/kWh)	13.5560	14.5704	7.5%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N70 network access charge will increase relative to N70's first and second block energy charges. Endeavour Energy has increased prices to the second block energy price relative to the first block energy price to better signal the cost of large residential consumers on Endeavour Energy's network. Increases in energy rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs.

6.2.2 Residential time of use (type 5) – N705

The following table provides the proposed prices for the residential time of use tariff (type 5) for 2012/13. N705 is an optional residential time of use tariff with less than 0.1% of residential customers charged using this tariff.

Table 6.2 – Proposed 2012/13 residential time of use (type 5) tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	314.76	346.75	10.2%
Peak Energy Charge (c/kWh)	18.2825	19.4021	6.1%
Shoulder Energy Charge (c/kWh)	10.7396	11.1583	3.9%
Off Peak Energy Charge (c/kWh)	4.3502	4.7280	8.7%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N705 network access charge will increase relative to the energy based charges. Consistent with Endeavour Energy's network tariff strategy, peak DUOS energy rates have increased relative to shoulder and off peak DUOS energy rates. However, the impact of TCR and CCF tariff increases has resulted in a greater percentage increase in the off peak energy price at the NUOS pricing level. This is due to the higher proportional representation of TCR and CCF tariffs in the off peak NUOS energy price relative to the peak and shoulder NUOS energy prices.

6.2.3 Residential time of use – N706

The following table provides the proposed prices for the residential time of use tariff for 2012/13. N706 is an optional residential time of use tariff with less than 0.1% of residential customers charged using this tariff.

Table 6.3 – Proposed 2012/13 residential time of use tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	183.00	200.75	9.7%
Peak Energy Charge (c/kWh)	18.2825	19.4021	6.1%
Shoulder Energy Charge (c/kWh)	10.7396	11.1583	3.9%
Off Peak Energy Charge (c/kWh)	4.3502	4.7280	8.7%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N706 network access charge will increase relative to the energy based charges. Consistent with Endeavour Energy's network tariff

strategy, peak DUOS energy rates have increased relative to shoulder and off peak DUOS energy rates. However, the impact of TCR and CCF tariff increases has resulted in a greater percentage increase in the off peak energy price at the NUOS pricing level. This is due to the higher proportional representation of TCR and CCF tariffs in the off peak NUOS energy price relative to the peak and shoulder NUOS energy prices.

6.2.4 General supply inclining block tariff – N90

The following table provides the proposed prices for the default general supply inclining block tariff for 2012/13. N90 is Endeavour Energy's primary general supply tariff with approximately 97.7% of general supply customers charged using this tariff.

Table 6.4 – Proposed 2012/13 general supply inclining block tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	157.38	182.50	16.0%
First Block Energy Charge (c/kWh)	8.9714	9.5425	6.4%
Second Block Energy Charge (c/kWh)	10.7674	11.4518	6.4%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N90 network access charge will increase relative to N90's energy based charges. The network access charge represents a minor portion of an average General Supply customer's network electricity bill, as demonstrated in chapter 7 of this proposal. Increases in energy rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs.

6.2.5 General supply time of use (type 5) – N845

The following table provides the proposed prices for the general supply time of use tariff (type 5) for 2012/13. Approximately 0.5% of general supply customers are charged using the N845 tariff.

Table 6.5 – Proposed 2012/13 general supply time of use (type 5) tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	435.54	456.25	4.8%
Peak Energy Charge (c/kWh)	14.3230	16.6180	16.0%
Shoulder Energy Charge (c/kWh)	9.1828	10.3236	12.4%
Off Peak Energy Charge (c/kWh)	6.2305	5.7870	-7.1%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N845 network access charge will decrease relative to N845's peak and shoulder energy charges.

Endeavour Energy is in the process of transitioning N845's energy based charges to align with the energy charges applicable to customers on the type 6 metered general supply time of use tariff (N84). Increases in energy rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs.

6.2.6 General supply time of use – N84

The following table provides the proposed prices for the general supply time of use tariff for 2012/13. Approximately 1.7% of general supply customers are charged using the N84 tariff.

Table 6.6 – Proposed 2012/13 general supply time of use tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	270.84	313.90	15.9%
Peak Energy Charge (c/kWh)	15.7865	16.6180	5.3%
Shoulder Energy Charge (c/kWh)	10.3015	10.7174	4.0%
Off Peak Energy Charge (c/kWh)	4.0558	4.7539	17.2%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N84 network access charge will increase relative to N84's energy charges.

The network access charge represents a minor portion of an average general supply customer's network electricity bill, as demonstrated in chapter 7 of this proposal. Endeavour Energy is in the process of transitioning N84's energy based charges to align with the energy charges applicable to customers on the type 5 metered general supply time of use tariff (N845). Increases in energy rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs.

6.2.7 Controlled load tariffs – N50 and N54

The following table provides the proposed prices for the controlled load 1 tariff for 2012/13.

Table 6.7 – Proposed 2012/13 controlled load 1 tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	14.64	14.60	-0.3%
Energy Charge (c/kWh)	0.6137	0.6033	-1.7%

All prices in the above table are exclusive of GST

The following table provides the proposed prices for the controlled load 2 tariff for 2012/13.

Table 6.8 – Proposed 2012/13 controlled load 2 tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	14.64	14.60	-0.3%
Energy Charge (c/kWh)	2.8404	2.8806	1.4%

All prices in the above table are exclusive of GST

Below average increases in controlled load energy rates reflect the low marginal network cost driven by energy consumed on this tariff. The lower prices for controlled load tariffs provide an incentive for customers to save money by electing to switch eligible equipment to the tariff, while at the same time reducing peak demand on the network. Increases in energy rates include underlying increases in transmission cost recovery tariffs.

6.3 Low Voltage Demand Tariff Class

6.3.1 Low voltage time of use demand – N19

The following table provides the proposed prices for the low voltage time of use demand tariff for 2012/13.

Table 6.9 – Proposed 2012/13 low voltage time of use demand tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	5,522.94	6,405.75	16.0%
Peak Energy Charge (c/kWh)	4.2524	4.7188	11.0%
Shoulder Energy Charge (c/kWh)	2.5169	2.8959	15.1%
Off Peak Energy Charge (c/kWh)	0.9142	1.1261	23.2%
High Season Peak Demand Charge (\$/kVA/Month)	14.2243	15.1507	6.5%
Low Season Peak Demand Charge (\$/kVA/Month)	13.3248	14.0852	5.7%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N19 network access charge will increase relative to the energy based charges at the DUOS level. This strategy is consistent with Endeavour Energy's network tariff strategy as outlined in section 5.3 of this document which, among other things signals Endeavour Energy's intention to increase peak DUOS energy rates relative to shoulder and off peak DUOS energy rates and increase peak demand rates to better signal the costs of peak demand on the distribution network. In 2012/13 however, the impact of TCR and CCF tariff increases has resulted in a greater percentage increase in the off peak and shoulder energy price at the NUOS pricing level. This is due to the higher proportional representation of TCR and CCF tariffs in the off peak and shoulder NUOS energy prices relative to the peak NUOS energy price.

The network access charge represents a minor portion of an average low voltage time of use demand customer's network electricity bill, as demonstrated in chapter 7 of this proposal. Peak DUOS demand charges increase relative to N19's energy charges to better reflect the costs of peak demand usage on Endeavour Energy's network. Increases in energy and demand rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs.

6.3.2 Transitional time of use – N89

The following table provides the proposed prices for the transitional time of use tariff for 2012/13.

Table 6.10 – Proposed 2012/13 transitional time of use tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	5,522.94	6,405.75	16.0%
Peak Energy Charge (c/kWh)	15.4888	16.1865	4.5%
Shoulder Energy Charge (c/kWh)	8.4023	8.8762	5.6%
Off Peak Energy Charge (c/kWh)	1.2261	1.5438	25.9%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N89 tariff is a transitional tariff designed to transition customers to the N19 (low voltage time of use demand) tariff. The proposed N89 tariff component increases are designed to manage this transition. Increases in energy rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs. In 2012/13, the impact of TCR and CCF tariff increases has resulted in a greater percentage increase in the off peak energy price at the NUOS pricing level. This is due to the higher proportional representation of TCR and CCF tariffs in the off peak NUOS energy price relative to the peak and shoulder NUOS energy prices. Increases in energy rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs.

6.4 High Voltage Demand Tariff Class

6.4.1 High voltage time of use demand – N29

The following table provides the proposed prices for the high voltage time of use demand tariff for 2012/13.

Table 6.11 – Proposed 2012/13 high voltage time of use demand tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	9,139.02	10,599.60	16.0%
Peak Energy Charge (c/kWh)	3.0747	3.3640	9.4%
Shoulder Energy Charge (c/kWh)	2.1781	2.3635	8.5%
Off Peak Energy Charge (c/kWh)	0.8254	0.9796	18.7%
High Season Peak Demand Charge (\$/kVA/Month)	10.2785	11.0383	7.4%
Low Season Peak Demand Charge (\$/kVA/Month)	9.6237	10.2848	6.9%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N29 network access charge will increase relative to the energy based charges at the DUOS level. This strategy is consistent with Endeavour Energy's network tariff strategy as outlined in section 5.3 of this document which, among other things signals Endeavour Energy's intention to increase peak DUOS energy rates relative to shoulder and off peak DUOS energy rates and increase peak demand rates to better signal the costs of peak demand on the distribution network. In 2012/13 however, the impact of TCR and CCF tariff increases has resulted in a greater percentage increase in the off peak and shoulder energy price at the NUOS pricing level. This is due to the higher proportional representation of TCR and CCF tariffs in the off peak NUOS energy price relative to the peak and shoulder NUOS energy prices.

The network access charge represents a minor portion of an average high voltage time of use demand customer's network electricity bill, as demonstrated in chapter 7 of this proposal. Peak demand charges increase relative to N29's energy charges to better reflect the costs of peak demand usage on Endeavour Energy's network. Increases in energy and demand rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs.

6.4.2 Individually calculated high voltage tariffs

The individually calculated high voltage NUOS tariffs have been provided to the AER on a confidential basis as Appendix E of this *2012/13 Annual Pricing Proposal* on the basis that the tariffs contain customer specific data.

6.5 Subtransmission Voltage Demand Tariff Class

6.5.1 Subtransmission time of use demand – N39

The following table provides the proposed prices for the subtransmission time of use demand tariff for 2012/13.

Table 6.12 – Proposed 2012/13 subtransmission time of use demand tariff

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	14,358.18	16,654.95	16.0%
Peak Energy Charge (c/kWh)	2.7917	3.0191	8.1%
Shoulder Energy Charge (c/kWh)	1.9546	2.0792	6.4%
Off Peak Energy Charge (c/kWh)	0.7675	0.9058	18.0%
High Season Peak Demand Charge (\$/kVA/Month)	7.8524	8.4971	8.2%
Low Season Peak Demand Charge (\$/kVA/Month)	7.3960	8.0023	8.2%

All prices in the above table are exclusive of GST

Endeavour Energy's proposed 2012/13 N39 network access charge will increase relative to the energy based charges at the DUOS level. This strategy is consistent with Endeavour Energy's network tariff strategy as outlined in section 5.3 of this document which, among other things signals Endeavour Energy's intention to increase peak DUOS energy rates relative to shoulder and off peak DUOS energy rates and increase peak demand rates to better signal the costs of peak demand on the distribution network. In 2012/13 however, the impact of TCR and CCF tariff increases has resulted in a greater percentage increase in the off peak and shoulder energy price at the NUOS pricing level. This is due to the higher proportional representation of TCR and CCF tariffs in the off peak NUOS energy price relative to the peak and shoulder NUOS energy prices.

The network access charge represents a minor portion of an average subtransmission time of use demand customer's network electricity bill, as demonstrated in chapter 7 of this proposal. Peak demand charges increase relative to N39's energy charges to better reflect the costs of peak demand usage on Endeavour Energy's network. Increases in energy and demand rates include underlying increases in transmission cost recovery and climate change fund recovery tariffs.

6.5.2 Individually calculated subtransmission voltage tariffs

The individually calculated subtransmission voltage NUOS tariffs have been provided to the AER on a confidential basis as Appendix E of this *2012/13 Annual Pricing Proposal* on the basis that the tariffs contain customer specific data.

6.6 Inter-Distributor Transfer Tariff Class

The inter-distributor NUOS tariffs have been provided to the AER on a confidential basis as Appendix E to this *2012/13 Annual Pricing Proposal* on the basis that the tariffs contain customer specific data.

6.7 Unmetered Supply Tariff Class

The following table provides the proposed prices for the unmetered streetlighting NUOS tariff for 2012/13.

Table 6.13 – Proposed 2011/12 unmetered streetlighting NUOS

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	0.00	0.00	0.0%
Energy Charge (c/kWh)	8.8698	9.5056	7.2%

All prices in the above table are exclusive of GST

The following table provides the proposed prices for the unmetered traffic signal NUOS tariff for 2012/13.

Table 6.14 – Proposed 2012/13 unmetered Traffic Signal NUOS

Charging Parameter	Existing NUOS Tariff 2011/12	Proposed NUOS Tariff 2012/13	% change
Network Access Charge (\$pa)	n/a	0.00	n/a
Energy Charge (c/kWh)	n/a	9.5425	n/a

All prices in the above table are exclusive of GST

6.8 Miscellaneous, monopoly and emergency recoverable works service charges

6.8.1 Miscellaneous service charges

Endeavour Energy's miscellaneous service charges are consistent with Appendix H of the *2009 Distribution Determination* and are shown in the following table.

Table 6.15 – Charges for miscellaneous services

Miscellaneous Service	\$
Special Meter Reading	44
Meter Test	73
Supply of Conveyancing Information – Desk Inquiry	37
Supply of Conveyancing Information – Field Visit	73
Off-peak Conversion	59
Disconnection Visit (Acceptable Payment Received)	44
Disconnection at Meter Box	88
Disconnection at Pole Top/Pillar Box	148
Rectification of Illegal Connection	221
Reconnection Outside Business Hours	95

All prices in the above table are exclusive of GST

6.8.2 Monopoly service charges

Endeavour Energy's monopoly service charges are consistent with Appendix H of the *2009 Distribution Determination* and are shown in the following table.

Table 6.16 – Monopoly service charges

Monopoly service	Underground urban residential subdivision (vacant lots)				Rural overhead subdivisions and rural extensions				Underground commercial and industrial or rural subdivisions (vacant lots – no development)				Commercial and industrial developments	Asset relocation or street lighting
Design information	Up to 5 lots			\$159	R2 per hour				R2 per hour				R2 per hour	R2 or R3 per hour (see para 1.4.2)
	6 to 10 lots			\$239										
	11 to 40 lots			\$398										
	Over 40 lots			\$478										
Design certification	Up to 5 lots			\$80	1 to 5 poles			\$80	Up to 10 lots			\$159	R3 per hour	R2 or R3 per hour (see para 1.4.2)
	6 to 10 lots			\$159	6 to 10 poles			\$159	11 to 40 lots			\$239		
	11 to 40 lots			\$239	11 or more poles			\$239	Over 40 lots			\$478		
	Over 40 lots			\$318										
Design rechecking	R2 per hour				R2 per hour				R2 per hour				R3 per hour	R2 or R3 per hour (see para 1.4.2)
Inspection of service work (level 1 work)	Grade	A	B	C	Grade	A	B	C	Grade	A	B	C	R2 or R3 per hour	R2 or R3 per hour (see para 1.4.2)
	per lot	per lot	per lot	per lot	per pole	per pole	per pole	per pole	per lot	per lot	per lot	per lot		
	First 10 lots	\$40	\$96	\$200	1 - 5 poles	\$48	\$96	\$176	First 10 lots	\$40	\$96	\$200		
	Next 40 lots	\$24	\$56	\$120	6 - 10 poles	\$40	\$80	\$159	Next 40 lots	\$40	\$96	\$200		
	Remainder	\$8	\$32	\$56	11+ poles (see para 1.4.2)	\$32	\$56	\$120	Remainder	\$40	\$96	\$200		
Access permit	Residential subdivisions: \$27 per lot combined fee				\$1181 maximum per access permit				\$1181 maximum per access permit				\$1181 maximum per access permit	\$1181 maximum per access permit
Substation commissioning					\$886 per substation (see para 1.4.2)				\$886 per substation (see para 1.4.2)				\$886 per substation (see para 1.4.2)	\$886 per substation (see para 1.4.2)
Administration	Up to 5 lots			\$193	Up to 5 poles			\$193	R1 per hour (max 6 hours)				R1 per hour (max 6 hours)	R1 per hour
	6 to 10 lots			\$258	6 to 10 poles			\$258						
	11 to 40 lots			\$322	11 or more poles			\$387						
	Over 40 lots			\$387										
Notice of arrangement	\$193													
Re-inspection (level 1 and 2 work)	R2 per hour (maximum 1 hour per level 2 reinspection)													
Re-inspection (service provider)	\$80 For the purpose of para 1.2(b), a DNSP may charge a fee that is less than this fee, but not a fee that is more than this fee.													
Access	R1 per hour													
Authorisation	\$159													
Inspection of service work (level 2 work)	All service connections: A Grade: \$20 per NOSW (NOSW = Notification of service work)				B Grade: \$33 per NOSW				C Grade: \$96 per NOSW					
Site establishment	\$139													

6.8.3 Emergency recoverable works

Endeavour Energy will levy charges for emergency recoverable works in a manner consistent with Appendix H of the 2009 *Distribution Determination* as provided below:

H.5.1 Changes for emergency recoverable works

- a. *The charge that a DNSP may levy for emergency recoverable works must not exceed the sum of the following:*
 - i. *110 per cent of the costs (other than labour costs) actually incurred in providing the emergency recoverable works and*
 - ii. *The cost of labour actually used to undertake the emergency recoverable works determined by applying 150 per cent of the R2 labour rate for that labour.*
- b. *For the avoidance of doubt, in the application of section H.5.1(a)(2), where a DNSP retains labour for a specified period for the purpose of that labour undertaking emergency recoverable works, the DNSP may only charge for so much of that specified period during which the labour actually undertakes the emergency recoverable works. For example, if a DNSP retains labour for a minimum specified period of four hours and the time required to actually undertake the emergency recoverable works is only one hour, the DNSP may only charge for the one hour and not the four hours.*

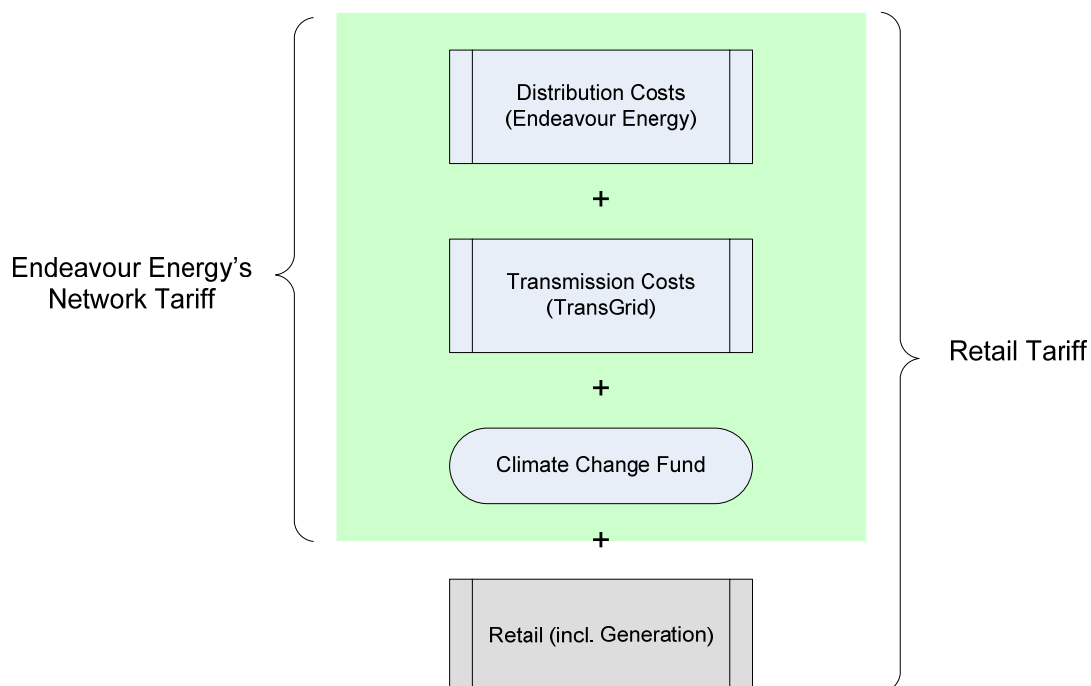
7

CUSTOMER IMPACTS

7.1 Endeavour Energy's Network Tariffs

Endeavour Energy's network use of system tariffs are an aggregation of distribution tariffs, transmission cost recovery tariffs and climate change fund recovery tariffs. Retailers generally pass through network tariffs to end use customers and add the costs of purchasing electricity from the wholesale market and other retail-related costs of selling electricity¹⁰ as illustrated in Figure 7.1 below. The customer impacts examined in this chapter relate only to network charges and do not include assumptions relating to retail charges.

Figure 7.1 – Costs recovered through Endeavour Energy's network tariffs

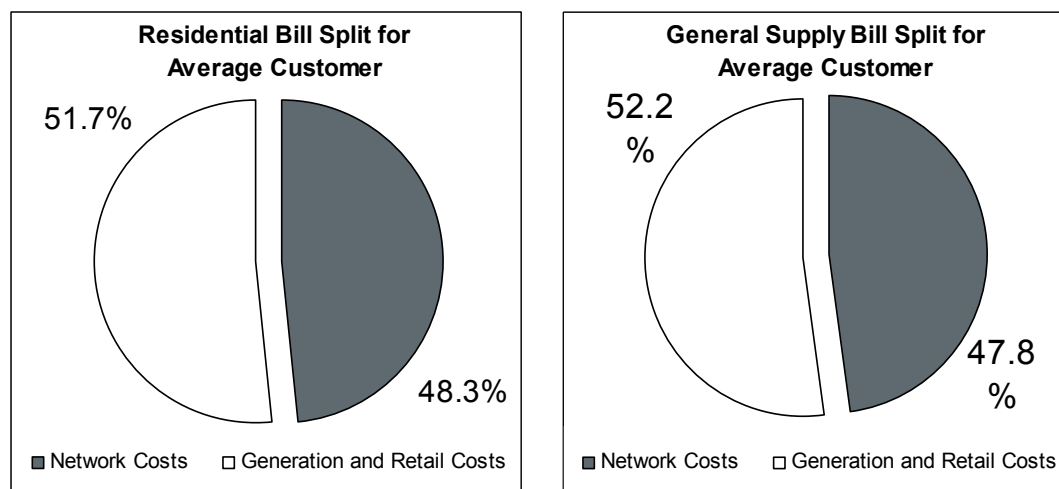


By way of example, figure 7.2 below provides the proportional network and retail components of an average regulated residential IBT and general supply IBT retail bill¹¹.

¹⁰ Including retail operating costs, retail margin and wholesale fees.

¹¹ Average regulated retail bills are calculated on the basis of the 2011/12 regulated Retail price for residential IBT and general supply IBT tariff customers in the Endeavour Energy network consuming 6,000kWh and 26,000kWh respectively.

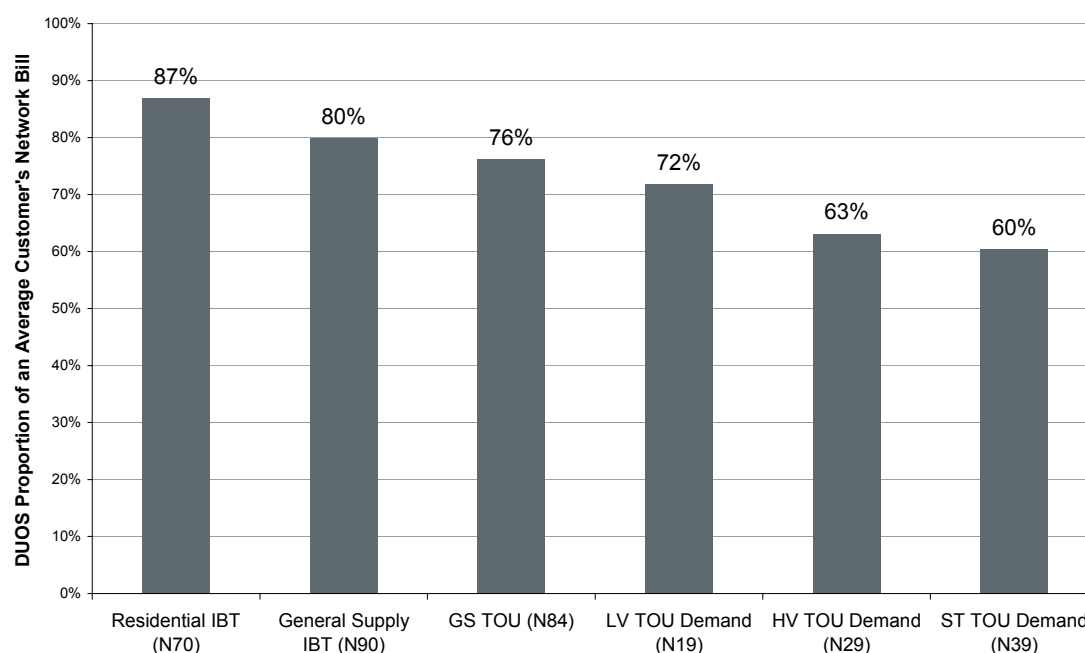
Figure 7.2 – Average regulated residential and general supply IBT bills by network and retail component – 2011/12



As demonstrated in figure 7.2 above the NUOS charges represent slightly less than 50% of the total electricity price in each case.¹²

Figure 7.3 below demonstrates that as average customer size increases and customers connect to higher voltages on the Endeavour Energy network the Distribution proportion of the total Network bill decreases and the proportional weighting of Transmission Cost Recovery and Climate Change Fund Recovery tariffs increases.

Figure 7.3 – Average DUOS proportion of a Network Bill – 2012/13



¹² Over 99% of Endeavour Energy's customers are charged for electricity on the basis of either residential IBT (N70) or general supply IBT (N90) network tariff. Not all customers are supplied on the basis of the regulated retail tariff.

The DUOS tariffs contained in the overall NUOS charges have been calculated in accordance with the weighted average price cap (WAPC) constraint as specified in the *2009 Distribution Determination*, adjusted for the outcomes of the Australian Competition Tribunal's merits review process. In nominal terms, the allowed increase in weighted average DUOS charges is 3.65%, as provided in Table 7.1 below:

Table 7.1 – Pricing compliance factors

Compliance Item	2012/13 Criteria
Consumer Price Index	3.39% ¹³
X Factor	-0.15%
D Factor	0.1% ¹⁴
WAPC $((1+CPI)*(1-X)*(1+D)-1)$	3.65%

Endeavour Energy's transmission cost recovery (TCR) tariffs are designed to recover transmission related costs, including TransGrid's transmission use of system (TUOS) charges, avoided transmission payments made to embedded generators and adjustments to balance Endeavour Energy's transmission overs and unders account. Endeavour Energy's transmission related costs are calculated to increase by 9.06% in 2012/13; with a portion of this increase to be offset by the operation of the transmission overs and unders account. The following table provides a break down of the drivers of the changes in Endeavour Energy's 2012/13 transmission costs.

Table 7.2 – Changes in 2012/13 transmission costs

Transmission Cost	2012/13 Change
A. Change in transmission related payments (a + b)	9.06%
- Impact of increase in transmission revenues payable to TransGrid (a)	7.86%
- Impact of increase in avoided TUOS payments to embedded generators (b)	1.20%
B. Change required to balance transmission overs and unders account	-0.57%
Total change in transmission costs $((1+A)*(1+B))-1$	8.44%

The customer impacts presented in the following section are calculated using Endeavour Energy's proposed NUOS tariffs and include increases in CPI, DUOS, climate change fund recovery and transmission cost recovery tariffs.

All customer impacts presented in this chapter arising from NUOS price changes are GST exclusive and have been calculated on the basis that historic customer energy consumption levels and patterns are indicative of 2012/13 consumption levels and patterns.

The distribution of impacts is provided where the number of customers on the tariff is sufficient and the data is available to provide a meaningful analysis.

¹³ Calculated in accordance with section 4.6 of the *2009 distribution determination*

¹⁴ As advised by the AER

7.2 Low Voltage Energy Tariff Class

7.2.1 Residential inclining block tariff – N70

The following table shows the expected network bill impacts of the proposed network price change for customers on the residential inclining block tariff.

Table 7.3 – Customer impact residential inclining block tariff

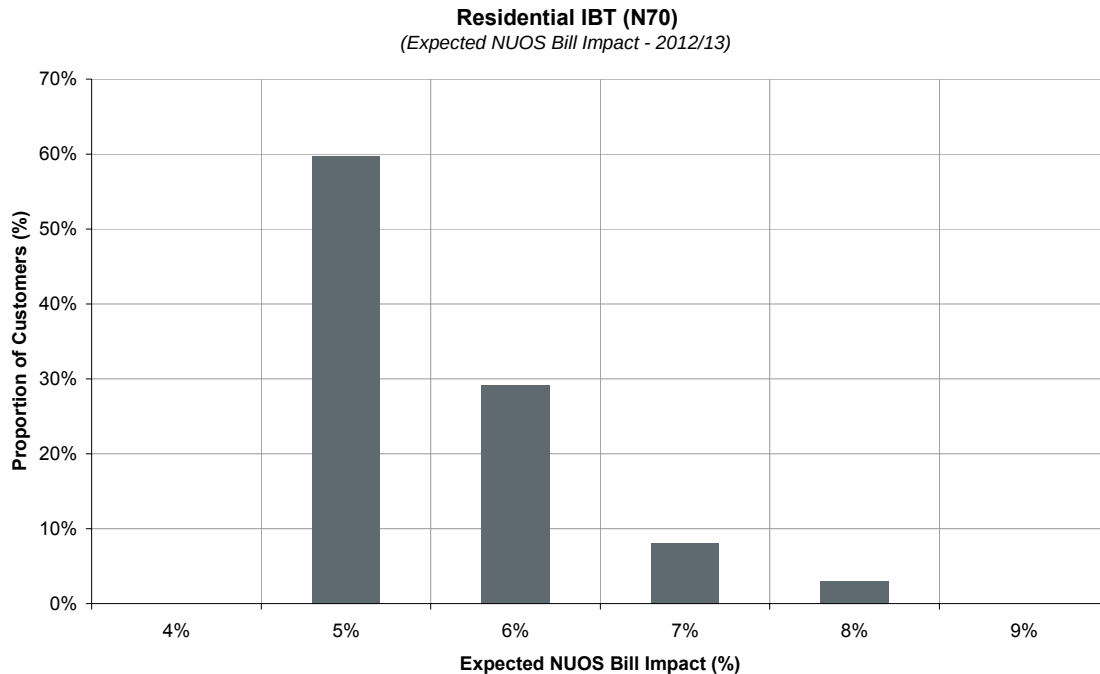
Annual Consumption	NUOS Bill (\$pa)		Change in NUOS Bill (%)
	2011/12	2012/13	
2,000	322.02	343.65	6.7%
4,000	530.58	559.56	5.5%
6,000*	739.14	775.46	4.9%
10,000	1,250.10	1,320.53	5.6%
15,000	1,927.90	2,049.05	6.3%

All indicative bill outcomes in the above table are exclusive of GST

() Approximate annual consumption of an average sized customer*

The following figure shows the impact distribution of the proposed network price change for customers on the residential inclining block tariff.

Figure 7.4 – Expected residential inclining block tariff NUOS bill impact distribution



7.2.2 Residential time of use (type 5) – N705

The following table shows the expected network bill impacts of the proposed network price change for customers on the residential time of use (type 5) tariff.

Table 7.4 – Customer impact residential time of use (type 5)

Annual Consumption	NUOS Bill (\$pa)		Change in NUOS Bill (%)
	2011/12	2012/13	
6,000	791.28	853.64	7.9%
10,000	1,108.96	1,191.56	7.4%
12,000*	1,267.80	1,360.52	7.3%
16,000	1,585.48	1,698.45	7.1%
20,000	1,903.16	2,036.37	7.0%

All indicative bill outcomes in the above table are exclusive of GST

() Approximate annual consumption of an average sized customer*

7.2.3 Residential time of use – N706

The following table shows the expected network bill impacts of the proposed network price change for customers on the residential time of use tariff.

Table 7.5 – Customer impact residential time of use

Annual Consumption	NUOS Bill (\$pa)		Change in NUOS Bill (%)
	2011/12	2012/13	
10,000	977.20	1,045.56	7.0%
30,000*	2,565.59	2,735.19	6.6%
50,000	4,153.99	4,424.81	6.5%
70,000	5,742.39	6,114.43	6.5%
90,000	7,330.78	7,804.06	6.5%

All indicative bill outcomes in the above table are exclusive of GST

() Approximate annual consumption of an average sized customer*

7.2.4 General Supply inclining block tariff – N90

The following table shows the expected network bill impacts of the proposed network price change for customers on the general supply inclining block tariff.

Table 7.6 – Customer impact general supply inclining block tariff

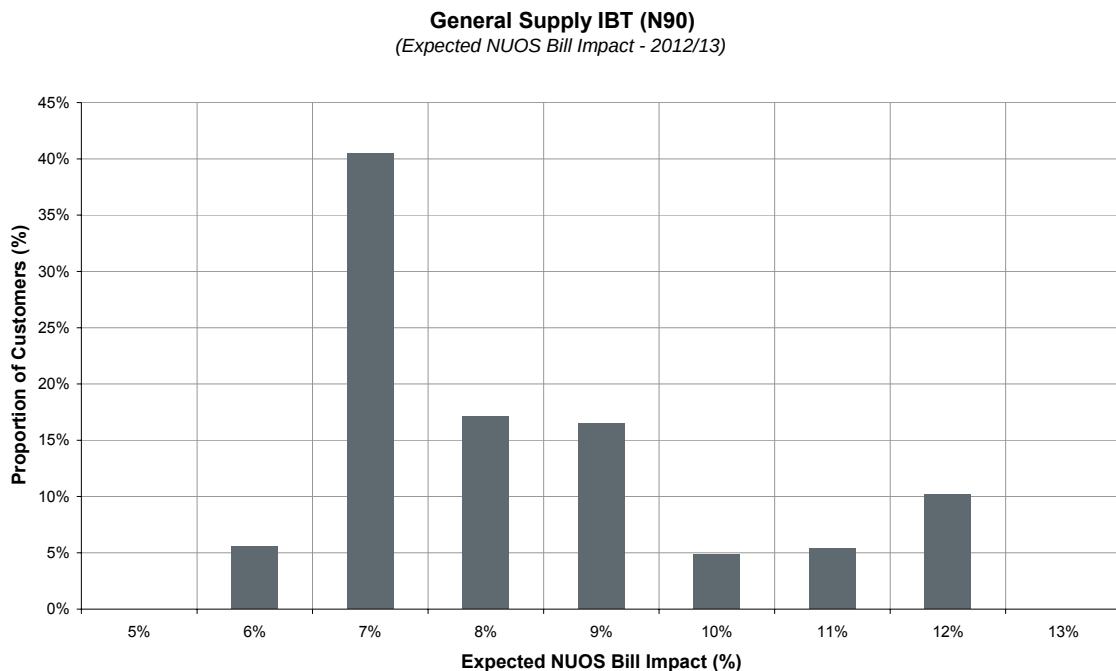
Annual Consumption	NUOS Bill (\$pa)		Change in NUOS Bill (%)
	2011/12	2012/13	
5,000	605.95	659.63	8.9%
10,000	1,054.52	1,136.75	7.8%
26,000*	2,777.30	2,969.04	6.9%
40,000	4,284.74	4,572.29	6.7%
60,000	6,438.22	6,862.65	6.6%

All indicative bill outcomes in the above table are exclusive of GST

() Approximate annual consumption of an average sized customer*

The following figure shows the impact distribution of the proposed network price change for customers on the general supply inclining block tariff¹⁵.

Figure 7.5 – Expected general supply inclining block tariff NUOS bill impact distribution

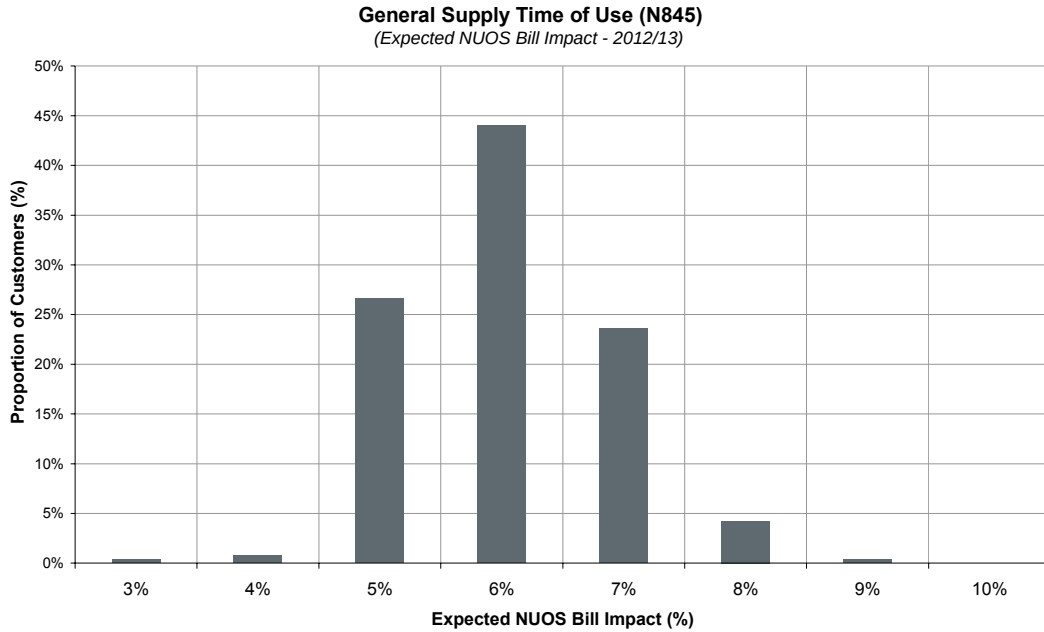


¹⁵ Customer distribution based on historic (09/10) data

7.2.5 General supply time of use (type 5) – N845

The following figure shows the impact distribution of the proposed network price change for customers on the general supply time of use (type 5) tariff.

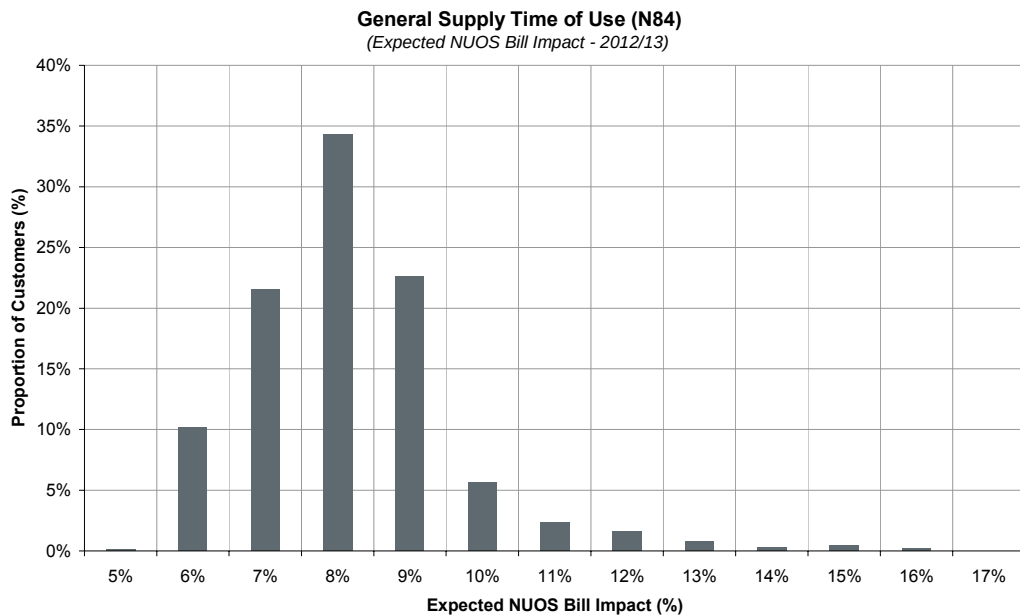
Figure 7.6 – Expected general supply time of use (type 5) NUOS bill impact distribution



7.2.6 General supply time of use – N84

The following figure shows the impact distribution of the proposed network price change for customers on the general supply time of use tariff.

Figure 7.7 – Expected general supply time of use NUOS bill impact distribution



7.2.7 Controlled load tariffs – N50 and N54

The following table shows the expected network bill impacts of the proposed network price change for customers on the controlled load 1 tariff.

Table 7.7 – Customer impact Controlled Load 1

Annual Consumption	NUOS Bill (\$pa)		Change in NUOS Bill (%)
	2011/12	2012/13	
1,000	20.78	20.63	-0.7%
3,000*	33.05	32.70	-1.1%
5,000	45.33	44.77	-1.2%
10,000	76.01	74.93	-1.4%

All indicative bill outcomes in the above table are exclusive of GST

() Approximate annual consumption of an average sized customer*

The following table shows the expected network bill impacts of the proposed network price change for customers on the controlled load 2 tariff.

Table 7.8 – Customer impact Controlled Load 2

Annual Consumption	NUOS Bill (\$pa)		Change in NUOS Bill (%)
	2011/12	2012/13	
1,000	43.04	43.41	0.8%
3,000*	99.85	101.02	1.2%
5,000	156.66	158.63	1.3%
10,000	298.68	302.66	1.3%

All indicative bill outcomes in the above table are exclusive of GST

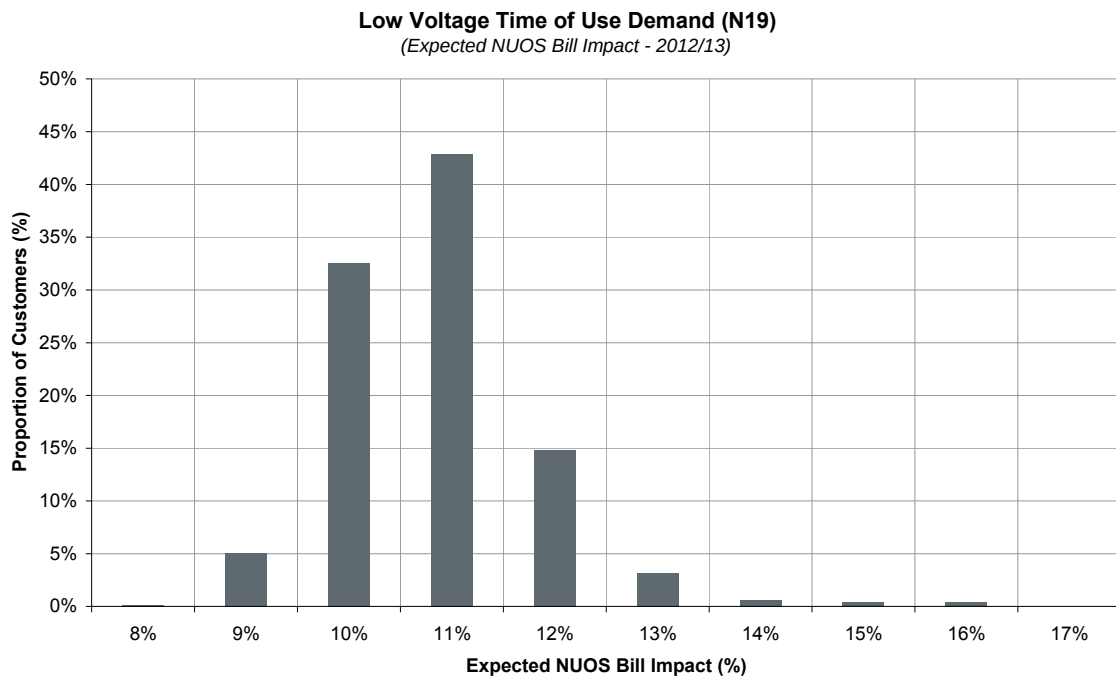
() Approximate annual consumption of an average sized customer*

7.3 Low Voltage Demand Tariff Class

7.3.1 Low voltage time of use demand – N19

The following figure shows the impact distribution of the proposed network price change for customers on the low voltage time of use demand tariff.

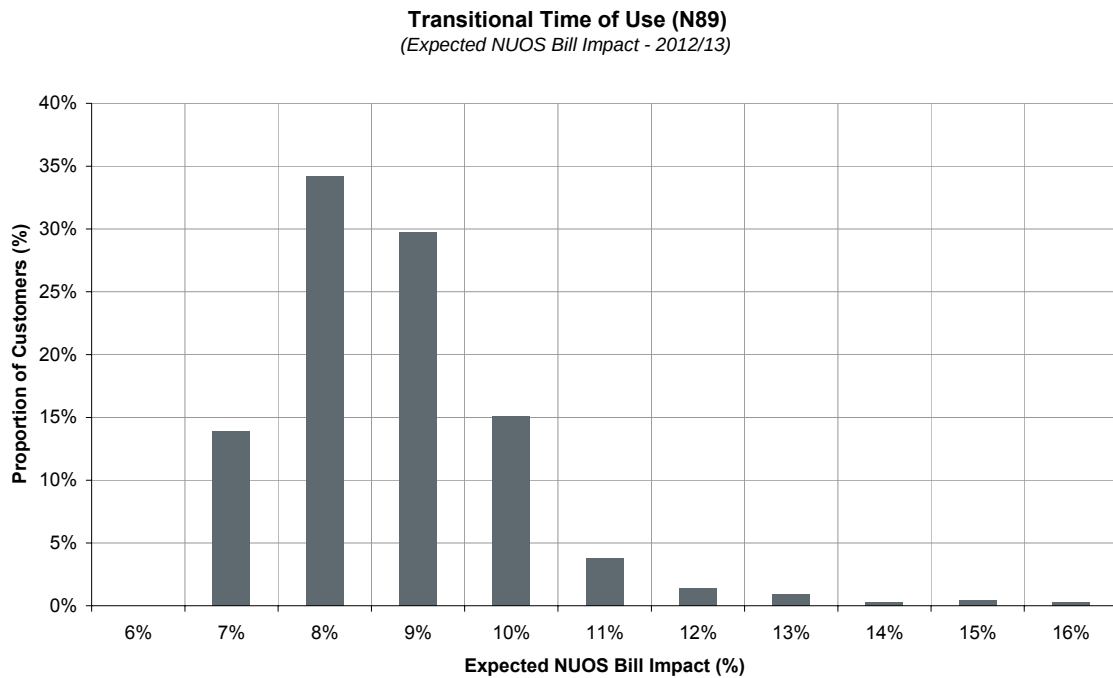
Figure 7.8 – Expected low voltage time of use demand NUOS bill impact distribution



7.3.2 Transitional time of use – N89

The following figure shows the impact distribution of the proposed network price change for customers on the transitional time of use tariff.

Figure 7.9 – Expected transition time of use NUOS bill impact distribution

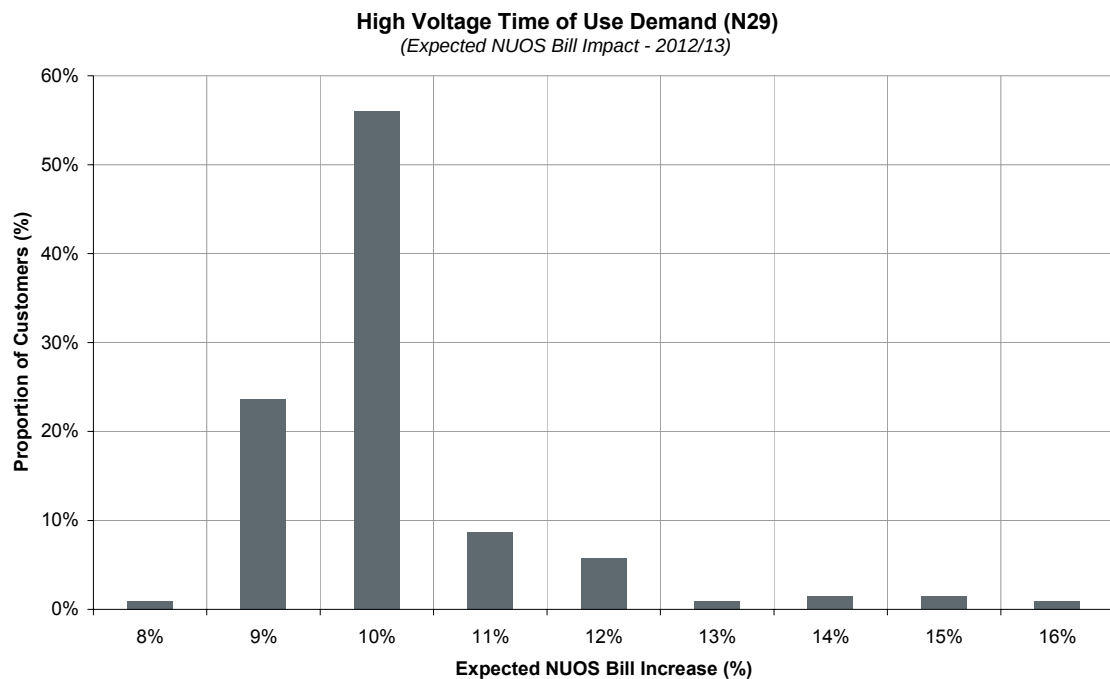


7.4 High Voltage Demand Tariff Class

7.4.1 High voltage time of use demand – N29

The following figure shows the impact distribution of the proposed network price change for customers on the high voltage time of use demand tariff.

Figure 7.10 – Expected high voltage time of use demand NUOS bill impact distribution

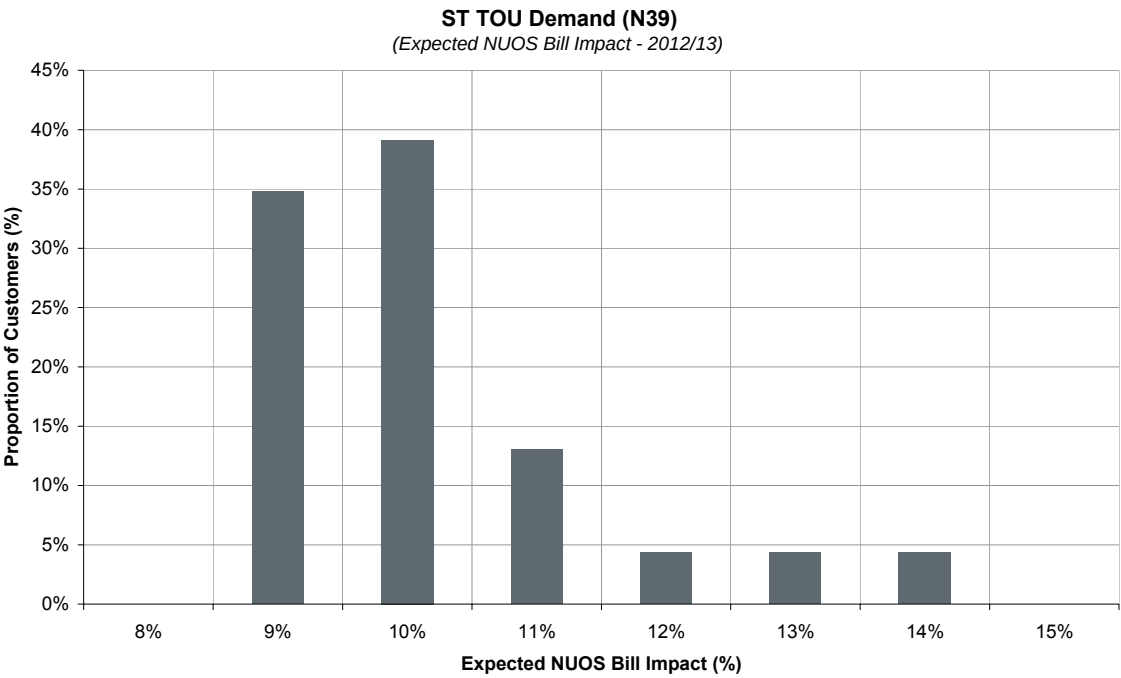


7.5 Subtransmission Voltage Demand Tariff Class

7.5.1 Subtransmission time of use demand – N39

The following figure shows the impact distribution of the proposed network price change for customers on the subtransmission time of use demand tariff.

Figure 7.11 – Expected subtransmission time of use demand NUOS bill impact distribution



7.6 Unmetered Supply Tariff Class

The following table shows the expected network bill impacts of the proposed network price change for customers on the unmetered streetlighting tariff.

Table 7.9 – Customer impact unmetered streetlighting tariff

Annual Consumption	NUOS Bill (\$pa)		Change in NUOS Bill (%)
	2011/12	2012/13	
1,000	88.70	95.06	7.2%
3,000	266.09	285.17	7.2%
5,000	443.49	475.28	7.2%
10,000	886.98	950.56	7.2%
20,000	1,773.96	1,901.12	7.2%

All indicative bill outcomes in the above table are exclusive of GST

Endeavour Energy is proposing the introduction of a new unmetered traffic control signal tariff from 1 July 2012. The new tariff will be applicable to traffic control signal lights owned and operated by the NSW Roads and Maritime Service. The new tariff has been developed to address the unique nature and extent of the customer's usage following an extensive consultation process with representatives of the NSW Roads and Maritime Service.

8

COMPLIANCE WITH REGULATORY REQUIREMENTS

8.1 2012/13 Compliance Requirements

The following table summarises the 2012/13 network pricing compliance criteria.

Table 8.1 – Pricing compliance factors

Compliance Item	2012/13 Criteria
Consumer Price Index	3.39% ¹⁶
X Factor	-0.15%
D Factor	0.1% ¹⁷
WAPC $((1+CPI)*(1-X)*(1+D)-1)$	3.65%

8.2 Compliance with the Weighted Average Price Cap

The following table demonstrates that Endeavour Energy's *2012/13 Annual Pricing Proposal* complies with the weighted average price cap (WAPC) X factor constraint outlined in the *2009 Distribution Determination* and based on the tariff classes outlined in Chapter 4 of this 2012/13 Annual Pricing Proposal. The table also satisfies clause 6.18.2(b)(4) of the *Transitional Rules*.

¹⁶ Calculated in accordance with section 4.6 of the *2009 distribution determination*

¹⁷ As advised by the AER

Table 8.2 – Compliance with the weighted average price cap

Tariff Class	Existing Weighted Average Revenue 2011/12 (\$'000)	Proposed Weighted Average Revenue 2012/13 (\$'000)	Change in Weighted Average Revenue (%)	Weighted Average Price Cap Constraint (%)
Low Voltage Energy	735,221	761,970	3.64%	
Low Voltage Demand	192,008	201,845	5.12%	
High Voltage Demand	44,731	46,133	3.14%	
Subtransmission Demand	33,259	32,644	-1.85%	
Inter-Distributor Transfers	6,612	6,609	-0.03%	
Unmetered Supply	9,639	9,871	2.41%	
Miscellaneous and Monopoly Fees	8,781	8,781	0.00%	
Total	1,030,251	1,067,855	3.65%	3.65%
Do the proposed DUOS tariffs meet the WAPC criterion?				Yes

8.3 Compliance with tariff class side constraints

The following table summarises the 2012/13 network tariff class side constraint criteria.

Table 8.3 – Pricing compliance factors

Compliance Item	2012/13 Criteria
Consumer Price Index	3.39 ¹⁸
X Factor	-0.15%
D Factor	0.1%
L Factor	2.00%
Tariff Class Side Constraint $((1+CPI)*(1-X)*(1+D)*(1+L)-1)$	5.72%

The following table demonstrates that Endeavour Energy's 2012/13 Annual Pricing Proposal complies with the tariff class side constraint outlined in the 2009 Distribution Determination.

¹⁸ Calculated in accordance with section 4.6 of the 2009 distribution determination

Table 8.4 – Compliance with tariff class side constraints

Tariff Class	Existing Weighted Average Revenue 2011/12 (\$'000)	Proposed Weighted Average Revenue 2012/13 (\$'000)	Change in Weighted Average Revenue (%)	Tariff Class Side Constraint (%)	Compliance with tariff class side constraint (yes/no)
Low Voltage Energy	735,221	761,970	3.64%	5.72%	Yes
Low Voltage Demand	192,008	201,845	5.12%	5.72%	Yes
High Voltage Demand	44,731	46,133	3.14%	5.72%	Yes
Subtransmission Demand	33,259	32,644	-1.85%	5.72%	Yes
Inter-Distributor Transfers	6,612	6,609	-0.03%	5.72%	Yes
Unmetered Supply	9,639	9,871	2.41%	5.72%	Yes

8.4 Pricing Principles

In accordance with 6.18.5 of the *Transitional Rules* Endeavour Energy must comply with the following pricing principles:

6.18.5 Pricing principles

(a) For each tariff class, the revenue expected to be recovered should lie on or between:

(1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and

(2) a lower bound representing the avoidable cost of not serving those customers.

(b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:

(1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and

(2) must be determined having regard to:

(i) transaction costs associated with the tariff or each charging parameter; and

(ii) whether customers of the relevant tariff class are able or likely to respond to price signals.

(c) If, however, as a result of the operation of paragraph (b), the Distribution

Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

8.5 Compliance with Avoidable and Stand Alone Cost Requirements

8.5.1 Definition of Avoidable and Stand Alone Cost

As part of its 2009/10 Initial Pricing Proposal, Endeavour Energy engaged CEG to provide an external review of Endeavour Energy's avoidable and stand alone cost calculation methodology and model.

As part of this review, CEG provided Endeavour Energy with two definitions of avoidable and stand alone cost.

Option1:

The first hypothesises the different costs associated with building a new network with and without the relevant customer class. This approach ignores the sunk nature of the existing network and calculates costs as follows:

- **Avoidable cost for tariff class A** - total cost of building and operating the network that Endeavour Energy would design to serve all customers less total cost of building and operating the network that Endeavour Energy would design to serve all customers except those in tariff class A; and
- **Stand alone cost for tariff class A** - total cost of building and operating the network that Endeavour Energy would design to serve only the customers in tariff class A.

Option 2:

The second is to acknowledge the existence of the network as it presently stands today and to calculate standalone/avoidable costs taking the existing network design as a given. Under this approach costs are calculated as follows:

- **Avoidable cost for tariff class A** - total cost to Endeavour Energy of using the existing network to serve all customers less total cost to Endeavour Energy of using the existing network to serve all customers except those in tariff class A; and
- **Stand alone cost for tariff class A** - total cost to Endeavour Energy of using the existing network to serve only the customers in tariff class A.

CEG note that the second definition (Option 2) involves a more 'real world' definition of avoidable costs. Namely, it is the costs that would actually be avoided if a set of customers did not need to be served (rather than the hypothetical costs that would have been avoided if that set of customers had never existed).

The *Transitional Rules* are not prescriptive about the methodology used to calculate avoidable and stand alone cost. However, given a regulatory framework which recognises the existence and value of sunk costs and does not, in general, seek to

conduct ex-post optimisation on the structure of the *Distribution Network Service Provider's* networks, one may reasonably conclude that the second method (Option 2) is consistent with the general approach set out in the *Transitional Rules*.

Consistent with the approach underpinning the 2009/10 and 2011/12 Annual Pricing Proposals submitted to the AER, Endeavour Energy has adopted the second method (Option 2) for this *2012/13 Annual Pricing Proposal*, as detailed in the following section.

For the purposes of the 2009/10 Initial Pricing Proposal, CEG was satisfied that Endeavour Energy's:

- Methodology for calculating avoidable and stand alone costs; and
- The model used to calculate avoidable and stand alone costs at the *tariff class*.

meet the requirements of section 6.18.5(a) of the *Rules*.

Endeavour Energy has maintained a consistent approach for this *2012/13 Annual Pricing Proposal*.

The CEG review of Endeavour Energy's avoidable and stand alone cost methodology and model is provided in Appendix J.

8.5.2 Avoidable and Stand Alone Cost Methodology

To estimate standalone costs, Endeavour Energy's *Regulatory Cost of Supply Model* makes an allocation of all costs (network and overhead costs) to each tariff class. Through this allocation, the stand alone cost of serving each separate tariff class must sum to the standalone cost of serving all customers. The implicit assumption underlying this calculation is that there are no economies of scale or scope in the provision of electricity networks. Since it is clear that there are economies of scale and scope and that they are material, CEG concluded for the purposes of the 2009/10 Initial Pricing Proposal that Endeavour Energy's stand alone costs will be materially higher than those estimated in its model.

Similarly, Endeavour Energy's model makes an allocation of all 'direct' (or causative) costs to estimate the level avoidable costs for each tariff class. These do not include overheads such as motor vehicles and information technology systems but do include a full allocation of all network assets and network operating and maintenance costs. Again, this assumes that there are no economies of scale or scope in relation to network assets (since each tariff class must bear a proportional share of direct costs with the proportions summing to unity). In effect, the only costs that are not treated as avoidable are overheads. Taking into account the existence of such economies of scale, CEG concluded for the purposes of the 2009/10 Initial Pricing Proposal that Endeavour Energy's avoidable costs will be materially lower than those estimated in its model.

Given that Endeavour Energy's proposed revenues fall between the bounds of avoidable and stand alone cost as estimated in its *Regulatory Cost of Supply Model*, and for the reasons set out above, Endeavour Energy considers that the approach meets the *Rule* requirements. That is, amendments to Endeavour Energy's estimates as suggested by CEG would cause these bounds to widen, rather than to narrow. Given that Endeavour Energy already estimates that its revenues fit within these 'too narrow' bounds, Endeavour Energy concludes that revenues would similarly fall within the broader bounds that would more accurately reflect stand alone and avoidable costs.

Accordingly, Endeavour Energy considers that its pricing meets the requirements of section 6.18.5(a) of the *Transitional Rules*.

8.5.3 Calculation of Avoidable and Stand Alone Cost

Table 8.3 – avoidable and stand alone cost calculation

Tariff Class	Expected DUOS Revenue (\$'000)	Avoidable Cost (\$'000)	Stand Alone Cost (\$'000)	Between Avoidable and Stand Alone Cost?
Low Voltage Energy	733,022	722,219	994,795	Yes
Low Voltage Demand	203,747	59,263	331,839	Yes
High Voltage Demand	46,704	20,693	256,949	Yes
Subtransmission Demand	32,849	16,087	195,859	Yes
Inter-Distributor Transfers	6,438	4,262	184,034	Yes
Unmetered Supply	10,281	1	272,577	Yes

Endeavour Energy's network tariff classes lie within the subsidy free range whereby the expected DUOS revenue collected from each tariff class lies between the avoidable and stand alone costs of supply and therefore meet the requirements of section 6.18.5(a) of the *Transitional Rules*.

8.6 Long run marginal cost

8.6.1 External Review of the Long Run Marginal Cost Methodology

In 2005, Endeavour Energy commissioned National Economic Research Associates (NERA) to develop a report into the Long Run Marginal Cost (LRMC) of Electricity Distribution in Endeavour Energy's network area.

In preparing its 2009/10 Initial Pricing Proposal, Endeavour Energy engaged CEG to provide an external review of Endeavour Energy's long run marginal cost methodology and model, both of which were developed on the basis of the NERA Report.

In that review CEG endorsed Endeavour Energy's:

- Methodology for calculating long run marginal cost; and
- The model used to calculate long run marginal cost by tariff *charging parameter*.

For the purpose of this *2012/13 Annual Pricing Proposal*, Endeavour Energy has adopted the same methodology and model for LRMC, as set out below.

The CEG review of Endeavour Energy's LRMC Methodology and Model is provided in Appendix K.

8.6.2 Marginal Cost Pricing

The principle of marginal cost plays a central role in the formulation of economically efficient prices. If prices are set below marginal cost, resources will be wasted in meeting customers' demands which cost more than the benefits customers derive from them; and

if prices are set above marginal cost, then they will discourage purchases by customers which would have been valued more than the cost to society of supplying them.

NERA¹⁹ notes that marginal costs are often differentiated according to whether they correspond to the short or long run. The distinction between these two categories has to do not with any specific length of time, but rather with the factors of production that are variable. The short run refers to a situation in which the investment in plant and equipment is fixed. Capacity can be neither added nor removed, although it can be used to a greater or lesser extent.

The short run marginal cost of electricity distribution is the cost to society of a customer using existing capacity in the network at any point in time. This is generally very low unless the system is capacity constrained²⁰, and reflects the fact that the great majority of the costs of an electricity network provider are fixed in the short run and do not vary with the usage of the network. However, when the system is capacity constrained the marginal costs of usage can be very high.

It should be noted that these costs do not generally reflect increased costs to the network service provider but rather reflect the costs to society of “customer A” using the network and thereby preventing “customer B” from using the network. That is, when the system is constrained the costs of one customer using the network are equal to the benefits foregone by other customers who are unable to use the network.

These benefits foregone are generally measured as the maximum amount a customer who is constrained would have been willing to pay for the last available unit of capacity. By definition, this is equal to the market-clearing price for existing capacity. That is, the price at which existing capacity is rationed amongst potential users. When capacity is not constrained, as noted above, the market-clearing price is zero and therefore the short run marginal cost of congestion is also zero.

The long run refers to a situation in which the investment in plant and equipment is variable. Existing firms may add or remove capacity; firms may enter or exit the market. A long-run marginal cost therefore indicates how costs change with respect to output or capacity when all factors of production including plant and equipment are variable. The long run marginal cost then will relate broadly to the annualised cost of augmenting capacity (in the case of electricity, at a particular voltage, at a particular location, at a particular time), generally, per unit of additional capacity provided (i.e., kW or kVA).

8.6.3 The importance of efficient pricing for network utilities

NERA²¹ note that the economies of scale or density applying to network utilities mean that supplying services at marginal cost may not be financially sustainable. This gives rise to a tension between economic efficiency and revenue adequacy. In such circumstances a firm differentiates its price so as to ensure:

¹⁹ NERA (2005) Long Run Marginal Cost of Electricity Distribution - A Report for Integral Energy p 8-9

²⁰ This abstracts from variable (demand related) operating and maintenance costs and, most significantly, the cost of electricity losses on the distribution system. The level of electricity losses imposed by a customer on the distribution system will also vary with the location of that customer in the network, the level of energy use by the customer and the level of capacity utilisation at the time of that energy use.

²¹ NERA (2006), Distribution Pricing Rule Framework, *Network Policy Working Group*, p 9

- That the total quantity produced is as close as possible to the quantity that corresponds to the intersection of demand and marginal cost, while maintaining overall financial viability; and
- That any differences between average cost and marginal cost are recovered from those customers who have the least responsive demand (with respect to price movements) referred to as price elasticity.

The above approach to pricing enables *DN*SP's to recover their average costs while maintaining incentives to invest in maintaining and augmenting their networks.

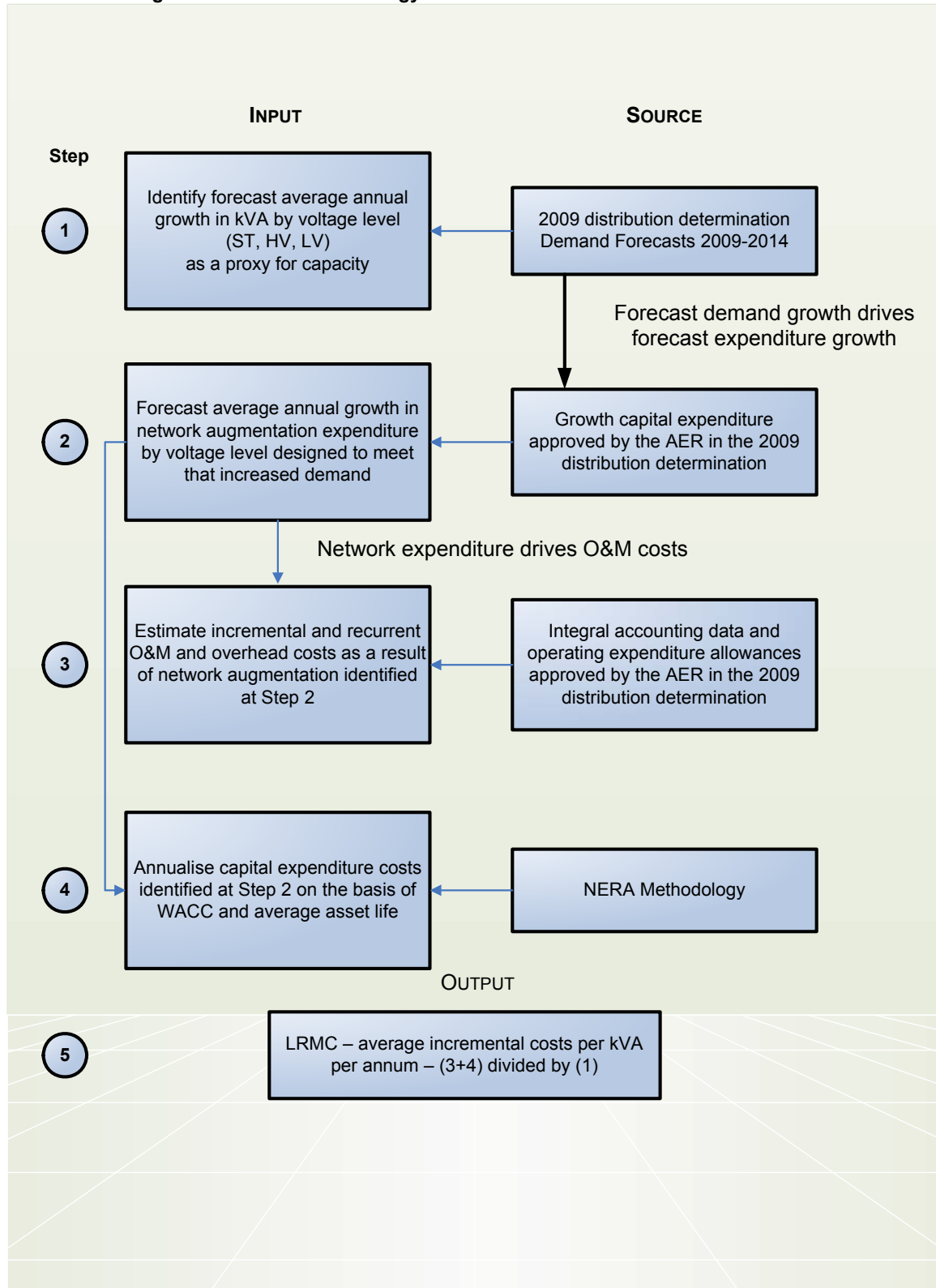
In this case it is important that one *charging parameter* of the tariff reflects the marginal cost of supply (however defined), providing effective signals for allocative efficiency of existing capacity; and for a second *charging parameter* to recover the remaining costs required to ensure that the network service provider can recover the fixed costs of the network. It is important in structuring the tariff that the second *charging parameter* be levied so as to impose the least possible distortion to customers' consumption behaviour.

8.6.4 Long Run Marginal Cost Methodology

Endeavour Energy's long run marginal costs are calculated on a dollar per kVA per annum basis at the relevant voltage level. The methodology uses the AER's 2009 *Distribution Determination* demand forecast, capital expenditure and operating expenditure allowances.

The following schematic illustrates Endeavour Energy's methodology as advised by NERA and previously endorsed by CEG, to arrive at an estimate of Endeavour Energy's LRMC. The process has been divided into five separate steps:

Figure 8.4: LRMC Methodology - National Economic Research Associates



The following LRMC estimates represent the cost associated with meeting forecast growth in critical peak demand for apparent energy (kVA). Critical peak demand is any level of demand that, should it occur often enough, would trigger an increase in network augmentation under Endeavour Energy's network planning.

Table 8.5 – Voltage level LRMC calculation

Voltage Level	LRMC Calculation (\$/kVA/pa)
Low Voltage	342.36
High Voltage	86.31
Subtransmission	58.18

It should be noted that the values for low voltage and high voltage already include the long run marginal cost of upstream (higher voltage) costs caused by consumption at the lower voltage level.

The formulas used to allocate the above LRMC by voltage into tariff *charging parameter* rates and the resultant LRMC tariffs by *charging parameter* have been provided to the AER on a confidential basis as Appendix B to this 2012/13 Annual Pricing Proposal.

Endeavour Energy considers that its 2012/13 Annual Pricing Proposal meets the LRMC requirements of section 6.18.5(b)(1) of the *Transitional Rules* as we take LRMC into account for each tariff *charging parameter* and use it as a guide for movements in our demand and peak energy pricing.

8.7 Transaction Costs

In accordance with clause 6.18.5(b)(2)(i) of the *Transitional Rules*, Endeavour Energy attempts to minimise any transaction costs arising from its network tariffs by limiting the complexity of the tariff structure and *charging parameters* within each tariff. The *charging parameters* applicable to each tariff are provided in Chapter 4 of this 2012/13 Annual Pricing Proposal while more detailed information of Endeavour Energy's tariffs and *charging parameters* are provided in the Network Price List available on the Endeavour Energy website.

Endeavour Energy has not introduced new tariff structures in 2012/13 and as such, transaction costs will not be increased as a result of the implementation of this 2012/13 Annual Pricing Proposal.

8.8 Response to Price Signals

In accordance with clause 6.18.5(b)(2)(ii) of the *Transitional Rules*, Endeavour Energy notes that efficiency gains of marginal cost pricing are realised when the tariff, based on the marginal cost of supply, induces a behavioural change in the customer²². To maximise this customer response Endeavour Energy signals the long run marginal cost of supply through those tariff *charging parameters* with the greatest price elasticity of demand, namely the variable consumption charges that are based on the customer's energy use and maximum demand.

Consistent with the principles of efficient pricing (section 8.6.3), and in accordance with clause 6.18.5(c) of the *Transitional Rules*, Endeavour Energy recovers the shortfall between marginal cost and average cost from those charging parameters with the lowest price elasticity of demand, namely the fixed *charging parameter* where possible, thus

²² NERA (2006), Distribution Pricing Rule Framework, *Network Policy Working Group*, p 31 - 35

minimising the distortion of the long run marginal cost signal inherent in Endeavour Energy's variable *charging parameters* and maximising the probability of efficiency gains realised through customer behavioural response to efficient price signals.

CLIMATE CHANGE FUND

9.1 Regulatory Requirement

In accordance with clause 6.18.2(b)(5A) of the *Transitional Rules*:

6.18.2 Pricing proposals

(b) A pricing proposal must:

(5A) in the case of a NSW Distribution Network Service Provider – set out how Climate Change Fund payments are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those amounts in any previous regulatory year.

(b1) In its first year pricing proposal after 1 July 2010 the NSW Distribution Service Provider must detail how the provider proposes to carry out adjustments to Climate Change Fund payments for the purposes of clause 6.18.2(b)(5A) and report to the AER on the recovery process under clause 6.18.7B(a) to (c).

6.18.7B Recovery of Climate Change Fund payments

- (a) In the case of a NSW Distribution Network Service Provider, a pricing proposal must provide for tariffs designed to pass on to customers Climate Change Fund payments.
- (b) The amount to be passed on to customers for a particular regulatory year must not exceed the estimated amount of Climate Change Fund payments for the relevant regulatory year adjusted for over or under recovery in any previous regulatory year.
- (c) The extent of the over or under recovery is the difference between:
 - (1) the actual amount of the Climate Change Fund payments in a previous regulatory year; and
 - (2) the amount passed on to customers in respect of Climate Change Fund payments in that previous regulatory year.

9.2 Climate Change Fund Requirement

On 18 April 2012, the NSW Government provided Endeavour Energy with advice that the Climate Change Fund contribution amount will most likely increase to \$74,250,000 in 2012/13. Consistent with NSW Government direction in previous years, Endeavour Energy has assumed that no more than 25% of the Climate Change Fund contribution can be recovered from residential tariffs.

9.3 Climate Change Fund Recovery Tariff Setting Methodology

Climate Change Fund recovery tariffs have been in place since 1 July 2005 and are levied on the energy (kWh) based *charging parameter* of tariffs only. Existing tariffs are annually adjusted such that the weighted average price change for the climate change fund recovery portion of network price is evenly applied to all tariffs to achieve the required annual contribution amount (subject to the 25% cap placed by the NSW Government on residential tariff contributions to the fund).

Endeavour Energy does not recover a contribution to the climate change fund from:

- controlled load tariffs as customers contribute to the fund through their primary tariff; or
- inter-distributor transfer tariffs as customers contribute to the fund through the tariffs offered by the destination distributor.

9.4 Climate Change Fund overs and unders account balance

In accordance with clause 6.18.2(b1) of the Transitional Rules, Endeavour Energy has adopted the unders and overs methodology used by Endeavour Energy to calculate the balance of the transmission overs and unders account. This methodology is consistent with the methodology outlined by the AER in Appendix I of the 2009 Distribution Determination for the unders and overs associated with transmission charges.

In accordance with clause 6.18.7B of the *Transitional Rules*, table 9.1 provides the forecast 2012/13 balance of Endeavour Energy's climate change fund overs and unders account.

Table 9.1 – Climate Change Fund overs and unders account balance (\$'000)

	2010/11 Actual	2011/12 Expected	2012/13 Forecast
Revenue from Climate Change Fund charges	44,823	43,636	74,593
Climate Change Fund Payments	44,513	44,684	74,250
Over / (Under) Recovery	310	(1,048)	343
Overs and Unders Account			
Annual Rate of Interest (applicable to balance)	10.02%	10.02%	10.02%
Semi Annual Rate of Interest	4.89%	4.89%	4.89%
Opening balance of trans overs/(unders) account	343	702	-327
Interest on opening balance	34	70	-33
Over / (under) recovery for financial year	310	-1,048	343
Interest on over/under recovery	15	-51	17
Closing Balance of Account	702	-327	0

Endeavour Energy's Climate Change Fund recovery tariffs have been provided to the AER on a confidential basis as Appendix C to this *2012/13 Annual Pricing Proposal* as they contain individual customer data.

10

TRANSMISSION COST RECOVERY TARIFFS

10.1 Transmission Costs

Endeavour Energy's transmission cost recovery (TCR) tariffs are designed to recover transmission related costs, including TransGrid's transmission use of system (TUOS) charges, avoided transmission payments made to embedded generators and adjustments to balance Endeavour Energy's transmission overs and unders account. Endeavour Energy's transmission related costs are calculated to increase by 9.06% in 2012/13; however, this increase is partially offset by the operation of the transmission overs and unders account. The following table provides a break down of the drivers of the changes in Endeavour Energy's 2012/13 transmission costs.

Table 10.1 – Change in 2012/13 transmission costs

Transmission Cost	2012/13 Increase
A. Increase in transmission related payments (a + b)	9.06%
- Impact of increase in transmission revenues payable to TransGrid (a)	7.86%
- Impact of increase in avoided TUOS payments to embedded generators (b)	1.20%
B. Increase required to balance transmission overs and unders account	-0.57%
Total increase in transmission costs $((1+A)*(1+B))-1$	8.44%

10.2 Regulatory Requirement

10.2.1 Transitional Rule Requirements

In accordance with clause 6.18.7 of the *Transitional Rules*, and with respect to the recovery of charges for transmission use of system (TUOS) services, Endeavour Energy's *2011/12 Annual Pricing Proposal* must satisfy the following criteria:

6.18.7 Recovery of designated pricing proposal charges

(a) A pricing proposal must provide for tariffs designed to pass on to customers the designated pricing proposal charges to be incurred by the Distribution Network Service Provider.

(b) The amount to be passed on to customers for a particular regulatory year must not exceed the estimated amount of the designated pricing proposal charges adjusted for over or under recovery in accordance with paragraph (c).

(c) The over and under recovery amount must be calculated in a way that:

(1) subject to subparagraphs (2) and (3) below, is consistent with the method determined by the AER in the relevant distribution determination for the Distribution Network Service Provider;

(2) ensures a Distribution Network Service Provider is able to recover from customers no more and no less than the designated pricing proposal charges it incurs; and

(3) adjusts for an appropriate cost of capital that is consistent with the rate of return used in the relevant distribution determination for the relevant regulatory year.

(d) Notwithstanding anything else in this clause 6.18.7, a Distribution Network Service Provider may not recover charges under this clause to the extent these are:

(1) recovered through the Distribution Network Service Provider's annual revenue requirement;

(2) recovered under modified clauses 6.18.7A(a) to (c) or clause 6.18.7B; or

(3) recovered from another Distribution Network Service Provider.

Where:

Designated pricing proposal charges are defined under Chapter 10 of the Rules as:

Any of the following:

(a) charges for designated pricing proposal services;

(b) avoided Customer TUOS charges;

(c) charges for distribution services provided by another Distribution Network Service Provider, but only to the extent those charges comprise:

(1) charges incurred by that Distribution Network Service Provider for designated pricing proposal services; or

(2) charges for standard control services;

(d) charges or payments specified in rule 11.39.

Designated pricing proposal services are defined under Chapter 10 of the Rules as:

Any of the following services:

(a) prescribed exit services;

(b) *prescribed common transmission services; and*

(c) *prescribed TUOS services.*

10.2.2 2009 Distribution Determination Requirements

In its *2009 Distribution Determination*, the AER stated its requirements with respect to the reporting of TUOS under and over recoveries. The AER decided that, in accordance with clause 6.12.1(19) of the *Transitional Rules* Endeavour Energy must submit, as part of its *pricing proposal*, a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with Appendix I of the AER's *2009 Distribution Determination*.

Endeavour Energy's approach to the recovery of TUOS costs has been developed to meet the requirements of the *Transitional Rules* and the AER's *2009 Distribution Determination* and to preserve TransGrid's transmission pricing signals where possible. Endeavour Energy's approach to the setting of Transmission Cost Recovery Tariffs and the transmission use of system overs and unders account balance, developed in accordance with Appendix I of the AER's *2009 Distribution Determination*, are discussed below.

10.3 Transmission cost recovery tariff methodology

The key principles of Endeavour Energy's Transmission Cost Recovery Tariff (TCR) Methodology are:

- Total TUOS allocated to network tariffs are aligned with the total estimated transmission charge to be paid by Endeavour Energy²³, adjusted for any overs and unders account balance;
- Transmission charges are allocated to network tariffs in a manner that reflects the cost drivers present in transmission pricing;
- Customers on an individually calculated tariff have transmission charges allocated in a manner that preserves the location and time signals of transmission pricing in accordance with the principles in Part J of chapter 6A of the *Rules*; and
- Network tariffs for smaller customer classes have transmission charges allocated on an energy basis, as location signals cannot be preserved in all cases due to metering limitations.

10.4 Transmission use of system overs and unders account balance

The following extract from Appendix I of the *2009 Distribution Determination* outlines Endeavour Energy's requirements with respect to the calculation of the transmission use of system overs and unders account balance:

²³ Transmission charges for 2012-13 based on Transmission Price List applicable to Endeavour Energy as advised by TransGrid on 2 April 2012.

Appendix I: Transmission use of system overs and unders account

To demonstrate compliance with clause 6.18.7 of the transitional chapter 6 rules and this final decision for the next regulatory control period, the AER requires the NSW DNSPs to maintain a transmission use of system (TUOS) overs and unders account. The NSW DNSPs must provide information on this account to the AER as part of their annual pricing proposals under clause 6.18.2(b)(7) of the transitional chapter 6 rules.

As part of their pricing proposals for each regulatory year of the next regulatory control period, the NSW DNSPs must provide the amounts for the following entries in their TUOS overs and unders account for the most recently completed regulatory year, the current regulatory year and the next regulatory year:

- 1. opening balance for each year*
- 2. interest accrued on the opening balance for each year, calculated at the rate of the post-tax nominal rate of return as approved by the AER in its distribution determination, or the equivalent nominal rate of return approved by IPART for the 2004–09 regulatory control period*
- 3. the amount representing the revenue recovered from TUOS charges applied in respect of that year, less the amounts of all transmission related payments made by the DNSP in respect of that year*
- 4. an adjustment to the net amount in item 3 by six months of interest, accrued at the approved nominal rate of return*
- 5. summation of the above amounts to derive the closing balance for each year.*

The NSW DNSPs must provide details of calculations in the format set out in table I.1 of this final decision.

In accordance with Appendix I of the 2009 *Distribution Determination*, and consistent with the example TUOS calculation provided in Table I.1, table 10.2 provides the forecast 2012/13 balance of Endeavour Energy's transmission use of system overs and unders account.

Table 10.2 – Transmission overs and unders account balance (\$'000)

	2010/11 Actual	2011/12 Expected	2012/13 Forecast
Revenue from TCR charges	205,681	191,745	216,851
Transmission Related Payments			
- Transmission charges to be paid to TNSP's	194,008	198,063	213,769
- Avoided TUOS payments	6,072	1,868	4,271
Total transmission related payments	200,080	199,931	218,040
Over / (Under) Recovery	5,601	(8,186)	(1,189)
Overs and Unders Account			
Annual Rate of Interest (applicable to balance)	10.02%	10.02%	10.02%
Semi Annual Rate of Interest	4.89%	4.89%	4.89%
Opening balance of trans overs/(unders) account	2,690	8,834	1,134
Interest on opening balance	270	885	114
Over / (under) recovery for financial year	5,601	(8,186)	(1,189)
Interest on over/under recovery	274	(400)	(58)
Closing Balance of Account	8,834	1,134	0

Endeavour Energy's transmission use of system cost recovery tariffs have been provided to the AER on a confidential basis as Appendix D to this *2012/13 Annual Pricing Proposal* as they contain customer specific information.

CUSTOMER REASSIGNMENT

11.1 Regulatory Requirement

In accordance with clause 6.18.4 of the *Transitional Rules*, Endeavour Energy is governed by the following customer reassignment principles:

6.18.4 Principles governing assignment or re-assignment of customers to tariff classes and assessment and review of basis of charging

(a) In formulating provisions of a distribution determination governing the assignment of customers to tariff classes or the re-assignment of customers from one tariff class to another, the AER must have regard to the following principles:

(1) customers should be assigned to tariff classes on the basis of one or more of the following factors:

(i) the nature and extent of their usage;

(ii) the nature of their connection to the network;

(iii) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement;

(2) customers with a similar connection and usage profile should be treated on an equal basis;

(3) however, customers with micro-generation facilities should be treated no less favourably than customers without such facilities but with a similar load profile;

(4) a Distribution Network Service Provider's decision to assign a customer to a particular tariff class, or to re-assign a customer from one tariff class to another should be subject to an effective system of assessment and review.

Note:

If (for example) a customer is assigned (or reassigned) to a tariff class on the basis of the customer's actual or assumed maximum demand, the system of

assessment and review should allow for the reassignment of a customer who demonstrates a reduction or increase in maximum demand to a tariff class that is more appropriate to the customer's load profile.

(b) If the charging parameters for a particular tariff result in a basis of charge that varies according to the usage or load profile of the customer, a distribution determination must contain provisions for an effective system of assessment and review of the basis on which a customer is charged.

11.2 Proposed Compulsory Re-Assignment of Customers from 1 July 2012

Endeavour Energy intends to compulsorily assign 76 customers from their existing tariff to a new tariff in 2012/13 under two separate programs.

Under the first program, Endeavour Energy intends to compulsorily assign 75 customers with annual consumption in excess of 160 MWh pa from the general supply IBT, general supply time of use and transitional time of use tariffs to the low voltage time of use demand and transitional network tariff from 1 July 2012.

The customers targeted for re-assignment meet the following criteria:

- Have an annual consumption in excess of 160MWh pa;
- Have time of use metering capable of supporting the proposed tariff; and
- The expected network bill impact of the reassignment to the destination tariff is less than the bill impact the customer would have received had they stayed on their existing tariff.²⁴

This reform proposal is consistent with Endeavour Energy's pricing policy of compulsory demand pricing for all customers with annual consumption greater than 160 MWh pa, which was introduced after significant public consultation on 1 July 2004.

Under the second program, Endeavour Energy intends to compulsorily assign a single customer from the general supply IBT to the newly created traffic control signal tariff. The new tariff has been developed to address the unique nature and extent of the customer's usage following an extensive consultation process with the customer.

A summary of the proposed compulsory re-assignment of customers is provided in the following table:

²⁴ Bill impacts calculated on the basis of the 2010/11 annual consumption volume by customer.

Table 11.1 – compulsory customer assignment

Origin Tariff	Proposed Tariff	Customers Assigned
General Supply IBT	Transitional Time of Use	14
	Low Voltage time of use Demand	54
	Unmetered Traffic Light Tariff (New in 2012/13)	1
General Supply Time of Use	Transitional Time of Use	2
	Low Voltage time of use Demand	2
Transitional Time of Use	Low Voltage time of use Demand	3
Total		76

In accordance with Appendix J of the *2009 distribution determination* the AER requires that Endeavour Energy submit reasonable estimates of 2010/11 quantities (t-2) for each origin tariff that the customer is currently on, and for the tariff that Endeavour Energy intends to move the customer.

To be eligible for compulsory tariff re-assignment Endeavour Energy requires that the customer has metering installed that is capable of supporting the proposed tariff and that 2010/11 quantities are available in the form of the destination tariffs (ie peak, shoulder and off peak energy and demand tariff charging parameters). The reasonable estimates contained in this *2012/13 Annual Pricing Proposal* employ the customer's actual metered data by the proposed tariffs charging parameters in the 2010/11 financial year.

Upon approval of the this *2012/13 Annual Pricing Proposal*, Endeavour Energy will satisfy the procedures for reassigning customers to tariff classes as determined by the AER in Appendix A of the *2009 distribution determination*. Endeavour Energy will write to the customer's retailer, who acts on the customers behalf, informing them of the proposed tariff reassignment.

The notification letter provides the retailer with:

- The reasons for the reassignment;
- The criteria by which the customer was identified;
- The opportunity to object to the reassignment prior to its actioning; and
- Notification that an alternate dispute resolution process is available should the retailer be dissatisfied with Endeavour Energy's proposal.

12

ALTERNATE CONTROL SERVICES PUBLIC LIGHTING TARIFFS

12.1 Determination

The AER's final decision on Public Lighting tariffs and charges (Section 17.11) as it applies to Endeavour Energy is as follows:

In accordance with clause 6.12.1(12) of the transitional chapter 6 rules, the control mechanism for Endeavour Energy's alternative control services is:

- *a schedule of fixed charges in the first year of the next regulatory control period for assets constructed before 1 July 2009 and a schedule of fixed prices in the first year of the next regulatory control period for assets constructed after 30 June 2009*
- *a price path for the remaining years of the next regulatory control period.*

The schedule of fixed charges applicable to Endeavour Energy for assets constructed before 1 July 2009 is contained in appendix P. The price path that has been applied to develop these charges is a straight-line smoothing which provides a fixed indexation rate for each year of the next regulatory control period.

The schedule of fixed prices for 2009–10 applicable to Endeavour Energy for assets constructed after 30 June 2009 is contained in appendix R. The prices will be adjusted annually by the December quarter CPI data as published by the ABS.

12.2 Prices for Public Lighting Assets Constructed Before 1 July 2009 (Tariff Classes 1 and 2)

The AER's compliance mechanism for public lighting requires Endeavour Energy to submit, "as part of its pricing proposal, the total annual charge it proposes to levy on each of its public lighting customers over the next regulatory year" for public lighting assets constructed before 1 July 2009.

In our 2010/11 Pricing Proposal, Endeavour Energy made a number of adjustments to the charges originally determined by the AER and published in Appendix P of the 2009 *Distribution Determination*. These adjustments were made to address the revised WACC arising from the Australian Competition Tribunal's merits review decision and to

address an error in the original Determination where the raw prices used were in \$2009/10 when they should have been escalated to \$2010/11.

Incorporating these changes

To incorporate these changes, Endeavour Energy adopted the following approach to determine the Tariff Class 1 and 2 charges, as approved by the AER in 2010/11 and 2011/12.

The changes were incorporated into the AER model to generate a revised unsmoothed AER revenue. To smooth this revenue path the revenue for 2009/10 has been fixed at the level determined by the AER in its original *2009 Distribution Determination* and the revenue then smoothed over the four years from 2011/12 to 2013/14 to produce a Revised AER Revenue.

The Revised AER Revenue (smoothed) is then divided by the AER asset volumes (by tariff) to produce a per unit tariff which when multiplied by the AER volumes per public lighting customer generates the prices detailed in Confidential Appendix F.

Detailed unit pricing for Public Lighting Tariff Classes 1 and 2 is provided in Appendix G.

These prices are then consistent with those set out in Appendix P of the AER's final decision adjusted for the subsequent variation in applicable WACC arising from the Australian Competition Tribunal's merits review decision and a CPI escalation in base prices from \$2008/09 to \$2009/10.

In its modelling the AER assumed a reduction in the number of Tariff Class 1 and 2 lights installed which has not been achieved. Endeavour Energy proposes to recover the Tariff Class 1 and 2 charges by application of the tariffs generated by the AER volumes to actual inventories, consistent with historic billing methodologies and the method used for AER Public Lighting Tariffs 3 and 4.

The calculation of Endeavour Energy's 2012/13 Type 1 and 2 Public Lighting tariffs has been provided to the AER on a confidential basis as Confidential Attachment 4 – Public Lighting Model (Tariff Classes 1 and 2).

12.3 Prices for Public Lighting Assets Constructed After 30 June 2009 (Tariff Classes 3 and 4)

Endeavour Energy has adopted the methodology approved by the AER in 2011/12 to determine the prices for Tariff Class 3 and 4 for 2012/13.

These new per unit tariffs were then escalated by 3.39% being the December quarter CPI data as published by the ABS.

The 2012/13 prices proposed for public lighting assets constructed after 30 June 2009 are contained in Appendix H of this *2012/13 Annual Pricing Proposal*. The 2012/13 prices proposed for public lighting assets constructed after 30 June 2009 are consistent with those contained in Appendix R of the AER's final decision adjusted for the subsequent variation in applicable WACC and the CPI escalation.

Endeavour Energy proposes to apply these tariffs directly as inventories of the relevant components build up from 1 July 2009 onwards.

Endeavour Energy also notes the AER's comments in their *2009 Distribution Determination* regarding recovery of tax liabilities on AER Tariff 4 assets, and notes that it is our intention to only apply the tax component where Endeavour Energy takes ownership of the asset without having funded the capital cost. In cases where the customer funds the capital cost and ownership remains with the customer Endeavour Energy would not seek to recover the taxation component.

The calculation of Endeavour Energy's 2012/13 Type 3 and 4 Public Lighting tariffs has been provided to the AER on a confidential basis as Confidential Attachment 5 – Public Lighting Model (Tariff Classes 3 and 4).

GLOSSARY

Term	Definition
ABS	Australian Bureau of Statistics
AER	Australian Energy Regulator
Alternate Control Service	As defined in Chapter 10 of the Rules
Charging Parameters	As defined in Chapter 10 of the Rules
Connection	As defined in Chapter 10 of the Rules
Connection Point	As defined in Chapter 10 of the Rules
Connection Services	As defined in Chapter 10 of the Rules
CEG	Competition Economists Group
CPI	Consumer Price Index
Previous Regulatory Control Period	The regulatory period 1 July 2004 to 30 June 2009
Direct Control Services	As defined in Chapter 10 of the Rules
2009 Distribution Determination	New South Wales distribution determination 2009-10 to 2013-14 (AER, 28 April 2009)
Distribution Network Service Provider	As defined in Chapter 10 of the Rules
DUOS	Distribution Use of System
Distribution Use of System	As defined in Chapter 10 of the Rules
NUOS	Network Use of System
Pricing Proposal	As defined in Chapter 10 of the Rules
2009 Regulatory Control Period	The regulatory period 1 July 2009 to 30 June 2014
Regulatory Obligation or Requirement	As defined in Chapter 10 of the Rules
Regulatory Year	As defined in Chapter 10 of the Rules
Rules	National Electricity Rules
Standard Control Services	As defined in Chapter 10 of the Rules
Supply Point	The point of connection to Endeavour Energy's network
Tariff Class	As defined in Chapter 10 of the Rules
Transitional Rules	Clause 11.15 and Appendix 1 to Chapter 11 special transitional Chapter 6 arrangements for economic regulation of NSW distribution services for the <i>2009 regulatory control period</i>
WAPC	Weighted Average Price Cap